



Module 7

Running Mechanics

Running is an impact sport and therefore tends to have a relatively high injury rate in relation to other endurance sports. Because of this, a runner's form and, more specifically, the running gait are often analyzed for the purpose of improving efficiency and preventing injury.

It is important to reiterate that unless you are a physical therapist, chiropractor, or physician, nurse, physician assistant, etc., you cannot diagnose or treat a suspected injury. It is imperative that you always practice within your scope of knowledge and training. Dealing with injury is not within your scope of practice, and aside from telling a client to ice a sore area, the advice should always be to seek a qualified medical opinion.

Biomechanical assessments are the first place to start when evaluating your client. This will tell you a lot about areas that are potentially susceptible to injury or, at the very least, inefficient. The next phase is to assess clients while they are running.

Running Economy

The mechanics, and therefore the form of a runner, is often termed *running economy*. Many factors influence running economy, such as calf muscle girth, bone length, running form, and stride rate. Some factors are genetic in nature while others are not. Running economy relates to how efficient a runner is. A study found that the variation in efficiency between well-trained runners with good running economy and untrained runners with poor running economy is between 5 and 7 percent (715). It is likely that on an individual basis, the difference is even greater (705).

Just because an individual runs doesn't mean the person is efficient at it. Proper running form does not just happen. Running is a learned skill, just like other sports (731).

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Gait phases
- Ground reaction force
- Pronation and supination
- Q angle
- Aspects of proper running form
- Free energy
- Allocation of energy
- Hip-driven running/Spinal engine
- Midfoot strike versus heel strike
- Importance of the tibia position at foot strike
- Running-specific biomechanical assessments

❖ TERMINOLOGY

Pronation: Inward (medial) rotation of the foot. It is normal during a proper gait pattern for the foot to pronate. Foot pronation helps the body absorb stress from the impact of running and walking

- **Overpronation:** Excessive pronation

Supination (under pronation): Occurs when the foot “rolls” laterally. This places the majority of the weight on the outside aspect of the foot. Essentially, supination is a lack of pronation. During supination, the subtalar joint does not flex.

Heel-to-Toe Drop: Difference in height between the heel and forefoot of a shoe

Ground Reaction Force (GRF): Based on Newton’s third law, GRF is the force exerted by the ground on a body in contact with it (665).

Spinal Engine: Developed by Dr. Serge Gracovetsky, this model states that counter-rotation of the spine results in transverse hip/pelvic rotation and is a

major influence in the gait cycle. More specifically, Dr. Gracovetsky theorized that upon foot strike while running, energy was not absorbed by the ground but rather utilized up the body's kinetic chain to assist with locomotion.

Coupled Motion: This relates to a sequence of movements by different body parts to create an efficient and powerful movement.

Dorsiflexion: Pointing of the foot upward (flexion of ankle joint)

Plantar Flexion: Pointing of the foot downward (extension of ankle joint)

Free Energy: This relates to energy being stored (potential energy) via stretch of a muscle and/or connective tissue. By utilizing this free energy, the active muscle requirement is reduced while still producing the required amount of force.

Stretch-Shortening Cycle (SSC): Representative of an eccentric contraction of a muscle followed by a rapid concentric contraction of the same muscle. In respect to running, the quadriceps, obliques, and calves represent the muscles most influenced by the SSC.

Passive Movement: Representative of body movement that is not the result of active muscle contractions. Specific to running, lower leg (tibia/fibula) swing and hip flexion are several body movements that are influenced exclusively by passive movement.

Pedestrian Gait Model: The legs are the primary aspect of the body responsible for locomotion and the upper body is largely passive.

Windlass Mechanism: Prior to the foot landing when running, the toes are dorsiflexed, causing the plantar fascia to tighten, thus creating a stiff foot arch.

Vertical Oscillation: Vertical motion of a runner. While some vertical movement is normal and necessary during the running gait cycle, excessive vertical oscillation reduces the efficiency of a runner.

Torsion: Twisting of an object, especially from one end of an object relative to the other end.

Aponeurosis: Tendon that is typically flat and broad (i.e., sheet-like). Plantar fascia is an example of aponeurosis.

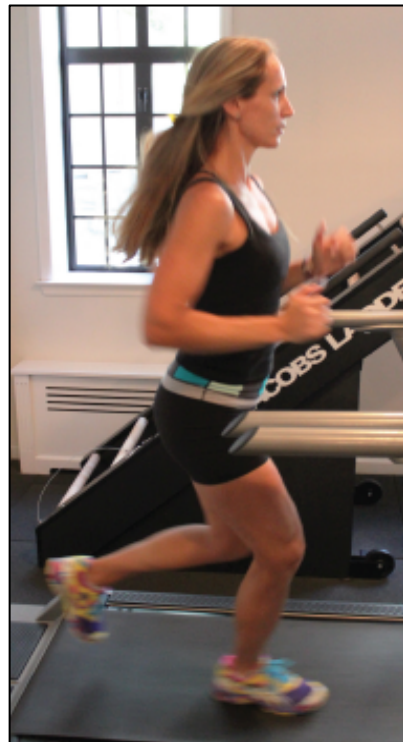
❖ GAIT PHASES

Running gait (cycle) has three phases:

1. **Recovery (Swing):** Any time a foot is in the air.
2. **Drive:** The drive phase starts when the leg is slightly behind the runner's hips and continues until right before the foot leaves the ground.
3. **Support:** The support (*stance phase*) starts when the foot hits the ground until the leg is directly underneath the hips. With a correct midfoot strike, the support phase is either eliminated or substantially minimized as the foot hits the ground under or just in front of the hips, thus beginning the drive phase.



Left Leg: Support **Right Leg:** Recovery
(aka Swing)



Right Leg: Drive **Left Leg:** Recovery
(aka Swing)

Figure 7.1 Gait phases

❖ GROUND REACTION FORCE (GRF)

Ground reaction force (GRF) is an important concept as it relates to the efficiency of a runner as well as the potential for injury. As noted below in the *Terminology* section, GRF relates to Newton's third law:

For every action, there is an equal and opposite reaction.

In regard to running, this means that there is an equal force acted upon the body in relation to the force of the foot strike on the ground. On average, each foot strike exerts a force 2.5 times one's bodyweight. This force is transmitted through your foot and up the leg. The form of a runner in relation to the person's biomechanics will influence the distribution of force on the musculoskeletal system (668).

Ground reaction forces change based on the gait of an individual. In a study by Keller et al., vertical GRF was evaluated during walking, slow jogging, and running (666). The study found that vertical GRF was 50 percent higher during slow jogging than walking or running. This increase in GRF was associated with greater vertical oscillation.

The higher the GRF, the greater the chance for injury (667). More specifically, an individual's specific biomechanics will likely influence the location and type of injury sustained (668).

❖ PRONATION / SUPINATION

PRONATION AND SUPINATION

There are three common designations of foot movement throughout the gait cycle.

1. **Mild Pronation**
2. **Overpronation**
3. **Supination (i.e., underpronation)**

These foot patterns are commonly evaluated statically in a standing position as well as dynamically during running. Since there is no contest in running events for how long one can stand statically, this is not an important benchmark. More specifically, it has been shown that foot and ankle mechanics vary greatly between standing and running (421, 422, 423).

A common assessment of pronation in a static position is the *wet test*. This assessment is performed by having an individual wet the bottom of the feet and then stand on something that absorbs water (i.e., a brown paper bag). By assessing the footprint left by the watermarks, the tester can then determine if the foot or feet are neutral, pronate slightly, overpronate, or supinate. As noted previously, the issue with this assessment is that foot and ankle mechanics are not the same when standing and running. More important, studies by the U.S. military demonstrated that assigning footwear based on one's footprint did little to reduce injury (432, 433).

Role of the Big Toe

Not all toes are created equal, at least not in terms of foot stabilization and forward propulsion. During running, the big toe (first metatarsophalangeal joint) plays a key role in the following:

- Stabilize the foot
- Regulate the degree of foot pronation
- Forward propulsion

The big toe in relation to the other toes is responsible for a much greater percent of foot and body stabilization as well as forward propulsion.

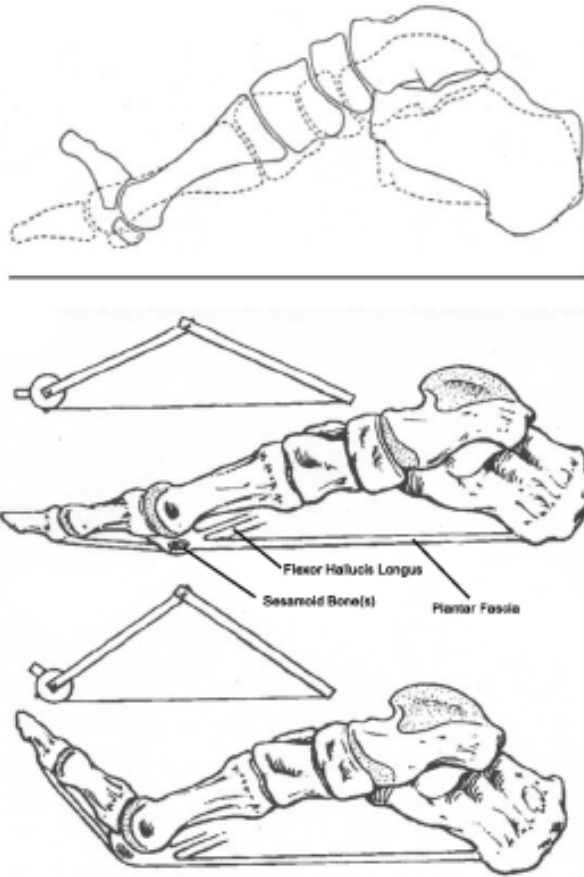


Figure 7.2 Windlass Mechanism (foot)

"PF-PlantarMove". Via Wikipedia-<http://en.wikipedia.org/wiki/File:PF-PlantarMove.jpg#mediaviewer/File:PF-PlantarMove.jpg>

The *windlass mechanism* of the foot pertains to the lifting (dorsiflexion) of the toes. As is seen in figure 7.2, the origin of the plantar fascia is the heel (calcaneus) and inserts at the head of the metatarsals.

Prior the foot landing when running, the toes are dorsiflexed, causing the plantar fascia to tighten, thus creating a stiff foot arch. This is representative of the *windlass mechanism* (611).

Upon landing the toes, plantar flex and the arch flattens (e.g., shock absorption).

When the leg moves into the drive phase, the toes begin to dorsi-flex. This action turns the arch into a rigid lever. This allows for a solid base of support to push off against.

As noted in figure 7.2, the sesamoid bones are two small (pea-size) bones that are embedded into a tendon (flexor hallucis brevis). The bones sit under the ball of the foot, at the big-toe joint. The sesamoid bones act as a fulcrum to provide the foot leverage when pushing off the ground (612).

First Metatarsophalangeal Joint Facts

- Carries 12 times more weight than the small toe
- Only toe made up of two bones as opposed to three
- It has a separate set of control muscles and tendon insertions than the rest of the toes.

Shoe Implications

Running shoes that are too narrow crowd the toes in the toe box. This negatively affects the aforementioned roles of the big toe, thus reducing the efficiency of a runner. A reduction in a runner's efficiency due to a cramped toe box likely increases a runner's metabolic cost. Therefore it is advised to run in shoes that have a wide enough toe box to allow the forefoot and toes to exist in their normal state (e.g., not cramped). Chronically wearing footwear with a crowded toe box can cause a bunion (biomechanical foot issue).

Shoe-Wear Patterns

If your client has an old pair of running shoes, looking at the wear pattern on the bottom of the soles will tell you a lot about the person's foot strike. For example, if there is very little or no wear on the heel, the client likely runs with a midfoot or forefoot strike.

Figure 7.3 denotes three common wear patterns and their associated foot/ankle mechanics.

If your client walks and runs in running shoes, the wear pattern will likely resemble that of a heel striker, even if the person runs with a midfoot strike. This is because when walking, the foot initially lands on the heel.

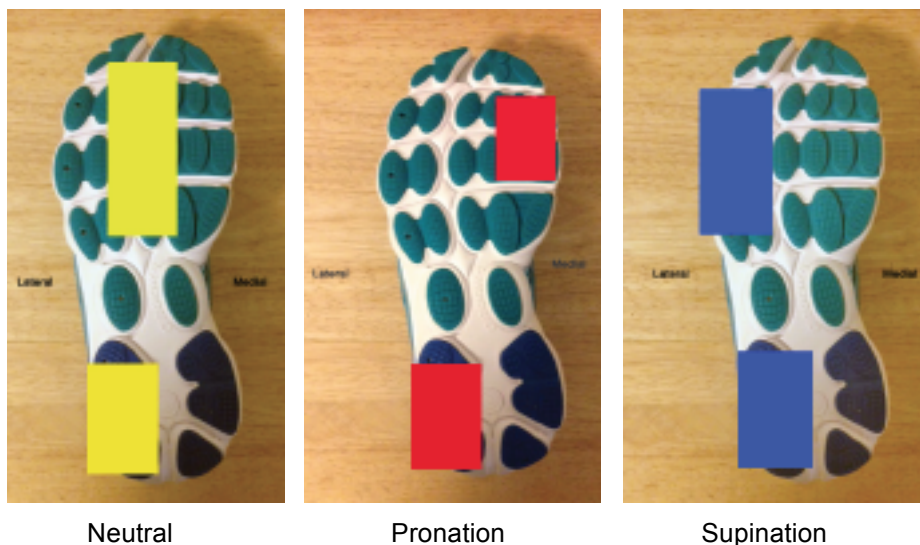


Figure 7.3 Shoe wear

Mild Pronation

Foot pronation refers to the inward (medial) rotation of the foot. It is normal during a proper gait pattern for the foot to pronate. Foot pronation helps the body absorb stress from the impact of running and walking. Many people assume that foot pronation is a bad thing, but as noted above, it is a natural and necessary movement.

The origin of pronation comes primarily from the ankle (subtalar and midtarsal joints). Below are the four primary phases of a normal gait cycle in regard to foot pronation:

1. **Foot strikes the ground:** slightly supinated (land on outside heel)
2. **Foot travels through the support phase:** Foot transitions from supinated to a pronated position.
3. **Foot begins the drive phase:** Foot is in a pronated position.
4. **Foot prepares to exit the drive phase and enter the recovery phase:** Foot begins to supinate. The big toe plays a large role in the stability of the foot, regulating the degree of foot pronation and forward propulsion.



Video 7.1 Mild foot pronation (rear foot)



Video 7.2 Mild foot pronation (front foot)

If you look at the worn sole of a pronator, typically the lateral aspect of the heel is worn because this represents the first place of impact, as the foot is supinated. As the foot rolls forward and moves into pronation, the middle to inside (medial) of the top third part of the sole tends to be the most worn. Normal pronation is also associated with compression of foot arch.

Overpronation

There is not a clear consensus on exactly how many degrees equal *overpronation*. However, many clinicians believe that a subtalar joint angle greater than 8–10 degrees equates to overpronation.

Overpronators typically have the same lateral heel wear pattern as someone who pronates, but because of the extreme medial roll of the foot, the wear on the top part of the shoe is almost exclusively on the medial side.

What Are Flat Feet?

Flat feet are often called *fallen* or *collapsed* arches. The medical term for flat feet is *pes planus*. Individuals with flat feet have little to no arch in the feet in the standing position. Those who overpronate often have flat feet, but this is not always the case.

A study by Menz (1998) found that static assessments to determine arch height classifications such as low or high arches were unreliable (610).

Orthotics and arch-strengthening exercises are often prescribed for those with flat feet.

The more a foot pronates, the more the femur and tibia internally rotate. If your client presents with excessive pronation that is accompanied with pain while running, it is advisable to have the individual meet with a specialist such as a podiatrist or physical therapist. Custom orthotics are often prescribed to help alleviate stress and pain caused by overpronation. Custom orthotics are made by taking a mold of an individual's feet. There are also over-the-counter orthotics that are not very costly. However, if you think your client would benefit from orthotics, it is best to refer the person to a professional. Motion-control running shoes are often recommended for runners who overpronate.

Overpronation is associated with increased ground reactive forces and foot-contact time (664).

A 2008 study by Cheung et al. found that the degree of foot pronation tended to increase with an increase in the distance ran (68). Weak hip abductor muscles (i.e., gluteus medius) can also contribute to overpronation.

Supination

Supination (also called *underpronation*) is essentially a lack of pronation. During pronation, the ankle joint (subtalar joint) flexes to allow medial rotation of the foot. During supination, the subtalar joint does not flex. As noted previously, the foot is

in a slightly supinated position right before striking the ground. For individuals who supinate, the foot stays in this position throughout the running gait cycle. This results in individuals “running” on the lateral aspect of their feet and pushing off the ground directly behind the small “pinkie” toe. This is evident if you look at the wear pattern on a shoe of someone who supinates. You will find that the outside part of the sole is worn, but not much else.

Those who supinate typically have a high and rigid foot arch. Based on the Cheung et al. study, this may or may not put the runner at increased risk for injury (194, 195, 196).

ASSESSING PRONATION AND SUPINATION

Like overpronation, there is not a clear consensus on exactly how many degrees constitute normal or mild pronation. Depending on the source or study, mild pronation ranges from 4–8 degrees of the subtalar joint angle (67).

Because of this lack of consensus, it is advantageous and practical to look for signs of excessive pronation/supination, as well make note of pain and/or injuries when assessing pronation and supination.

Visual Signs of Overpronation

- Knock-knees
- During the recovery phase, the lower leg swings excessively outward in relation to the upper leg. The foot also tends to turn outward (toe pointed laterally) at the rearmost part of the recovery phase. This often appears as if the foot briefly “flicks” outward.
- Substantial inward angle of the ankle

Visual Signs of Supination

- Bowlegged
- During the recovery phase, the lower leg swings excessively medially in relation to the upper leg
- Outward angle of the ankle

PRONATION IS NOT INDICATIVE OF PAIN OR INJURY

It is a popular belief that pronation is bad and will likely lead to injury. As noted earlier, pronation is a natural occurrence and acts to absorb stress, transfer the load on the feet, provide elastic energy, and facilitate a stable platform for the feet to push off of.

Within the running shoe world, there is often a drive to correct any foot/ankle articulation that is not considered a *neutral* or *normal* foot position – meaning a foot/ankle position that neither pronates nor supinates. Since most individuals' feet pronate to some degree, it is odd that a neutral foot position is viewed as “normal” (431). One could hypothesize that the driving force behind this are shoe manufacturers that stand to profit by selling shoes that “correct” foot/ankle movements such as pronation.

It is advantageous to not view pronation as being indicative of pain or injury, as study after study has proved this (424, 425, 426). In fact, several studies theorize that running injuries stem more from training errors (429) and hip dysfunction (430).

Q ANGLE

What Is a Q Angle?

In regard to overpronation, the term *Q angle* is often mentioned as a possible culprit. The Q angle is assessed by measuring the angle between two lines (57):

- **Red Line:** ASIS to the middle of the patella
- **Blue Line:** middle of patella to the tibial tubercle (also known as *tibial tuberosity*, it is the bump at top/middle of tibia, immediately below the patella)

In the following diagram, the Q angle is represented by the angle between the red and blue lines.

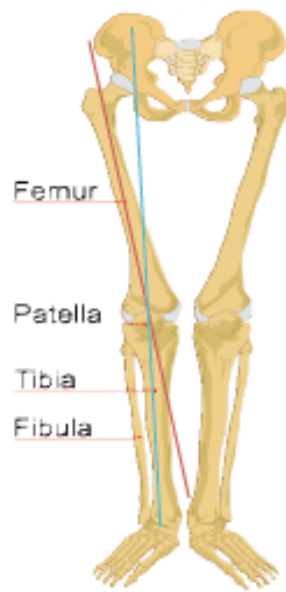


Figure 7.4 Q angle

Women typically have a greater Q angle than men because of a wider pelvis. The normal Q angle range for adults is 15–22 degrees, with women typically trending toward the larger angle (58, 59).

A large Q angle is often noted as a potential cause of patellofemoral syndrome (PFS). However, recent research has shown that there is not a strong correlation between an individual's Q angle and the incidence of PFS (646). Therefore a large Q angle should not be used to infer PFS.

KNEE VALGUS

This refers to the knee moving medially, often termed *excessive medial knee displacement (EMKD)*. During EMKD, both the femur and tibia medially rotate. This medial rotation is what causes the knee to move inward. Potential causes of EMKD are:

- Large Q angle
- Foot overpronation
- Overactive hip adductors
- Underactive hip abductors – primarily the gluteus medius
- Lack of ankle dorsiflexion

Of the aforementioned potential causes of EMKD, overpronation and underactive (i.e., weak) gluteus medius are the most cited.

Ankle Dorsiflexion

In a study by Fong et al., individuals with poor ankle dorsiflexion during foot impact (i.e., running) had greater ground reaction forces. Higher ground reaction forces are associated with EMKD (780). This is because ankle dorsiflexion is essentially a shock absorber and acts to absorb energy. A reduction in ankle dorsiflexion corresponds to a lack of energy absorption. This energy has to go somewhere and results in increased ground reactive force and likely EMKD.

❖ RUNNING FORM COMPONENTS

Ask a non figure skater to perform a triple axel on an ice rink and you'll likely get a look of bewilderment followed by, "*No freaking way!*" However, ask a nonrunner to go for a run and there is a strong likelihood that you'll get a, "*Sure, why not? Let's do it!*" While this is an extreme contrast, the point is that running is often viewed as a skill set that everyone knows how to do. While in its most basic element, this is likely correct. However, as a coach, you must understand that there is a massive difference between running and running correctly. Correct running form and proper running mechanics should not be viewed as a skill set that everyone naturally has but rather a learned skill that requires training to improve upon.

Following are 11 running-form principles based on biomechanics and running economy.



Figure 7.5 Proper running form

ENERGY ALLOCATION

To appreciate what the most important aspects of running form are, it is helpful to understand what elements of running form use the most energy.

The majority of energy goes to supporting body weight, followed by energy required for forward propulsion (drive phase of gait cycle) and energy required to swing the recovery leg forward. (585). Arm swing and balance have a minimal metabolic cost (586).

1 – Look Straight Ahead, Not Down or Up

When looking down, runners are not able to see obstacles or hazards that are in their path. The other primary reason for looking ahead is that the eyes cue the head position, which, in turn, cues the upper-body position. Runners who keep their eyes focused down tend to tilt their head downward, which causes the shoulders to roll anteriorly and rotate internally. Further down the kinetic chain, this can lead to excessive spinal flexion.

Conversely, looking upward or tilting the head backward cues the lower back to extend, which puts stress on the erector spinae muscles. The result of this arched back position (hyperextension) is that the feet strike the ground substantially in front of the body, thus lessening the efficiency of a runner.

2 – Upper Body Erect, Slight Forward Lean

An excessive anterior or posterior lean angle of the upper body results in excessive stress on the LPHC musculature. A correct slight forward lean does

not come from the pelvis, but rather from the ankles. From the heel to the shoulders, the body should have a slight forward lean when running. A good way to teach this is to have your clients stand with their feet in the frontal plane and then lean forward from the ankles. They will automatically move one leg forward to support themselves as their body moves forward.

It is common to see runners leaning back with substantial lumbar extension. This is especially true when a runner becomes fatigued.

Center of Mass (COM)

While running, one's center of mass is maintained over the base of support (i.e., legs/feet). By leaning slightly forward, one's COM is maintained over the base of support versus behind it. Balance is what enables a runner to control the COM.

As the body is in contact motion (vertical oscillation, planes: sagittal, frontal, transverse) when running, the COM is constantly changing.

A 2010 study by Hamner et al. found that the quadriceps and foot plantarflexors (gastrocnemius, soleus) are the primary contributors to accelerating the body's center of mass during running (695). This study, however, did not take into account the spinal engine model – noted later in this module.

Most people believe that the majority of the energy used while running is for forward propulsion. A study by Chang et al. found that 73 percent of a runner's total energy expenditure is used to support body weight and only approximately 20 percent is used for forward propulsion (730). This demonstrates the importance of proper form in regard to COM.

3 – No High (i.e., Bounce) or Long Stride

As with any exercise, efficiency is of utmost importance. A stride that pushes runners upward wastes energy that should be used to propel them forward. One way to cue this is to wear a hat with a brim and use the brim as a reference point. The brim of the hat should not rise or fall substantially. To further illustrate this concept, think about throwing a football. If you were asked to throw the ball 10 feet, you could throw it in a straight line and hit your target. However, if you were asked to throw it 75 feet, you would have to aim higher in the air to put an arc on

it. This is the same with a long stride versus a short stride. Longer strides require excessive vertical movement (inefficient), whereas short strides require very little vertical movement (efficient).

Vertical Oscillation

This relates to the degree of vertical motion of the body while running. As previously noted, the correct running gait will always have some degree of vertical oscillation. However, excessive vertical oscillation is inefficient. The more vertical oscillation one has, the more energy the person must expend to counteract gravity. This concept is discussed in greater detail later in this module.

4 – Foot Strikes the Ground Slightly in Front of or under the Body

Whether your client runs with a heel or midfoot strike, the correct form is for the foot to strike the ground directly under or slightly ahead of the hips. While more natural with a midfoot strike, it can also be accomplished with a heel strike. This relates directly to one's center of mass. There are two primary reasons why the foot should strike the ground under the body:

- 1. Hamstring Stress:** When the leg is excessively elongated in front of the body (termed *overstriding*) it puts substantial eccentric stress on the hamstrings.
- 2. Direct Power Transmission:** When the foot strikes the ground under the body, it immediately pushes down and rearward, thus propelling the runner forward.

Refer to video 7.6 for a dynamic visual representation of foot-strike location. The video contrasts heel- and midfoot-based strides and is representative of the foot striking the ground ahead of and under the body, respectively.

5 – Rotation of Pelvis (Hips)

Most individuals run with too little or too much pelvic rotation in the transverse plane. It is natural for the pelvis to rotate when walking/running, and a longer

stride length typically equates to increased pelvic rotation because of increased hip extension.

By eliminating pelvic rotation, you are effectively leaving out an important aspect of the body's natural kinetic chain.

Pelvic rotation helps to minimize the impact of running. Running with no pelvic rotation results in the feet hitting the ground and then moving rearward. This form lends itself to substantial impact on the legs and body (e.g., ground reaction force). When running with the correct amount of pelvic rotation, by the time the foot strikes the ground, that side of the pelvis is already rotating posteriorly, which reduces impact on the body (224). This is because the impact of the foot strike is dissipated slightly through the rotation of the pelvis.

While the pelvis needs to rotate, it should not over-rotate as this puts excessive pressure on the LPHC musculature and decreases the efficiency of the runner.

There are no hard numbers to go by in determining the correct amount of pelvic rotation. Therefore, visual observation is typically the most practical and efficient way to assess pelvic rotation and, more specifically, over-rotation. Running with excessive pelvic rotation requires the upper body to rotate substantially to counterbalance the pelvis and legs. If your client's torso rotates 45 degrees or more (noted at the shoulders) in the frontal plane, there is likely too much pelvic rotation. Fifteen to 30 degrees of upper-body rotation typically equates to the appropriate amount of pelvic rotation in the transverse plane.

As noted earlier, the degree of pelvic rotation is largely influenced by stride length and, more specifically, the amount of hip/leg extension.

When running with the correct amount of pelvic rotation, the hips and glutes drive the motion and the legs "*follow through*." This reduces the reliance on the hamstrings as the glutes and hip rotation initiate the drive phase. It could be theorized that those who run with static hips would be more susceptible to hamstring injury due to the minimized role of the glutes.

6 – Sweep, Don't Pound

This correlates with rotating the pelvis in the transverse plane while running. When the foot strikes the ground, it should already be moving in a sweeping motion toward the rear of the body. A good way to visualize this is a cat "pawing"

at the ground. As noted earlier, when the foot strikes the ground without traveling rearward simultaneously, it creates a pounding effect on the ground that decreases efficiency and likely increases the chance for injury. A sweeping foot strike creates a fluid and powerful stride, whereas a pounding foot strike creates a broken and nonfluid stride. Typically, runners who have a pounding foot strike have more vertical oscillation than those who sweep their feet.

Rotation of the pelvis in the transverse plane initiates and facilitates a sweeping foot strike.

7 – Keep Hands, Shoulders, and Face Relaxed

The goal of a runner is to have as little wasted energy as possible. When a runner's face is tightened, hands clenched, and shoulders shrugged (scapular elevation), the individual is diverting energy that should be going to the legs and core muscles.

8 – Arm Movement from Shoulders, Not Elbows

Arm movement should come primarily from the shoulders, not the elbows. Active flexion and extension of the elbows waste energy. Arm movement should not be forced, as arm motion does not drive the body forward. Arm motion is used to counterbalance momentum of the legs (695), and as such, excessive arm movement should not occur. If clients look like they are throwing uppercuts at Mike Tyson while running, they have too much arm movement!

Arms do not contribute substantially to propulsion when running (695). As just noted, the primary role of the arms is to counterbalance the momentum of the legs.

Another study found that the arms act to reduce torso and head rotation (696). The study also found that upper-body rotation is primarily a result of lower-body movement, not arm movement. A study by Kram et al. found that swinging the arms (not excessively) saved a runner approximately 4 percent in energy cost versus not swinging the arms. This is likely due to reduced muscular demand as a result of improved counterbalance and facilitating spinal counter-rotation via stretching of the latissimus dorsi (586).

Arm Crossover

There is a long-standing myth that when running the arms should not cross the vertical midline of the body (imaginary vertical line from the middle of nose to the center of the belly button). The most commonly cited reason for the arms not crossing the midline of the body is that it facilitates torso rotation. This is interesting because most runners lack the appropriate degree of transverse torso rotation. As a result of this flawed myth, the common advice is to run with the arms in a forward motion with no inward movement. Aside from looking very odd, this form puts excessive strain on the external rotator cuff muscles of the shoulders. Some degree of medial arm movement is suggested.

9 – Short Front Stride with Fast Leg Turnover

Typically, runners who land on the heel have a slower turnover than those who land midfoot. Regardless of the type of foot strike, the proper stride should focus on a relatively fast turnover with ample hip/leg extension. Shortening of one's stride pertains to eliminating or greatly reducing the foot strike ahead of the body, thus placing the foot strike closer to the center of mass. This is not intuitive for most runners and takes practice to learn. Conversely, leg/hip extension (leg movement behind the runner) typically should not be shortened unless a runner has too much extension. Stride length and hip rotation are discussed in greater detail later in this module.

While a short stride is typically favored over a long stride, too short of a stride can also be inefficient. This is largely because the runner is not taking full advantage of the body's stretch-shortening cycle (SSC). The ideal stride rate and stride length function on a bell curve (573).

10 – Hip Extension

Hip extension (e.g., leg behind body) is an essential aspect of proper running form. Individuals who lack sufficient hip extension because of tight hip flexors (iliopsoas) or a lack of transverse hip rotation often hyperextend their lumbar spine as a compensatory measure. Hip extension should come from the hip, not the lumbar spine. Proper hip extension is also a function of transverse pelvic rotation.

As noted later in the section discussing *passive movement*, proper hip extension allows for greater passive movement during the swing aspect of the gait cycle. This increases the efficiency of a runner.

ROLE OF THE ‘GLUTES’

Most runners will tell you that the glutes are some of the most important muscles for runners to develop – if not the most important – as this is where a runner’s power comes from. While this statement is largely correct, there are a few issues here.

- 1** – When most people refer to the “glutes,” they are referring to the gluteus maximus (GMax) – not the gluteus minimus or medius.
- 2** – While the GMax is critical for proper running form, other muscles are also very important to a runner’s form and performance.
- 3** – A runner’s form has to be correct to utilize the GMax correctly and adequately.

The primary role of the GMax is hip extension. While all runners utilize the GMax when running, most runners do not have the proper degree of hip extension either because of an extremely short posterior stride and/or little to no transverse hip rotation. As a result, the GMax is likely underactive.

In respect to running, the GMax is used concurrently to move the runner vertically and forward (upward diagonal motion). Too much vertical movement (i.e., vertical oscillation) results in wasted energy, increased impact on the body, decreased running economy, and a slower running speed. This form requires excessive GMax activation to move the runner vertically. Conversely, too little vertical oscillation results in a runner shuffling along.

If runners lack flexibility in their hip flexors, it is likely that they will lack adequate hip extension. This reduces the involvement of the GMax. Runners who have poor hip flexor range of motion often compensate via hyperextension of the lumbar spine.

Summary

The GMax is an important muscle for running, but it is utilized correctly only if a runner has the appropriate level of vertical oscillation and posterior hip extension.

11 – Neutral Spine

It is common to see runners with excessive posterior or anterior pelvic tilts. Of the two, running with an anterior pelvic tilt is more common. This often results in runners with a substantial arch (lordosis) in their lower back. As the hamstrings attach to the pelvis, if the pelvis is tilted too much anteriorly, it “pulls” on the hamstrings and creates excessive tension. As a result, during the swing phase of the gait cycle when the leg is in front of the body, the hamstrings are at a greater risk for injury because of increased tension under eccentric stress.

ASSESSING RUNNING FORM: TREADMILL vs. OUTDOORS

A common assessment for evaluating a runner’s form is to watch the person run on a treadmill. This is a valuable tool as it provides a controlled environment in which to assess a runner. However, one must be careful about what he or she is assessing. A treadmill is a good tool to assess foot/ankle position (i.e., pronation) and foot strike type. However, it has its weaknesses when assessing overall form.

When running outdoors, the foot pushes downward and rearward to propel a runner forward. However, when running on a treadmill, when the foot impacts the treadmill belt it is swept rearward. In other words, when running outdoors a runner must push off the ground, whereas when running on a treadmill, a runner must keep up with the treadmill belt. From a neuromuscular standpoint, these two running environments are different.

Many runners do not rotate their pelvis enough when running on solid ground. Conversely, when running on a treadmill, runners often have increased pelvic rotation due to the treadmill belt sweeping the legs rearward. Therefore running on a treadmill can artificially induce pelvic rotation that is not present when running on solid ground.

It is strongly advised to assess pelvic rotation, and a runner’s form in its entirety, while watching the person run outdoors at varying speeds.

Fatigue has been shown to alter the biomechanics of the lower extremities, specifically the knee and hips in the sagittal and transverse planes (337). This means that the starting form of your clients at the beginning of a run is likely not the same form that they will have at the end of the run. This may increase the chance for injury. It doesn't take a rocket scientist to appreciate that the form of runners at the start of a marathon is likely quite different from their form at the end!

Here is a summary of the benefits and drawbacks of treadmill-based assessments:

BENEFITS

- Easy to assess form because of the runner being stationary
- Can manipulate speed and incline
- Most gyms have them

DRAWBACKS

- A treadmill can induce transverse hip rotation that is not present when running outdoors.
- When running on a treadmill, the belt often facilitates the drive foot moving rearward versus the foot pushing rearward because of hip rotation/extension.

SELF-SELECTION OF RUNNING FORM

Some sources state that runners naturally self-select the most efficient running form/gait. This directly relates to a runner's economy and efficiency. So do runners naturally self-select the most optimal running cadence, foot strike, overall form, etc.? The answer is it depends.

For new runners, the answer is definitely no. Don't believe this? Go watch a marathon. Observe the form of elite/professionals as they run by. While they might not all have what is considered "textbook" form, they are very efficient as every body movement is purpose-driven to propel them forward with minimal wasted energy. Assuming that the form of elite runners is as close to perfect as one can get, compare this form with those of recreational runners in the rest of

the race. If the theory that all runners self-select the most efficient stride/form is correct, why would there be a difference in form between elites and recreational runners?

A 2014 study by De Ruiter et al. evaluated running efficiency based on stride rate (SR) and found that when self-selecting running SR, no novice runners selected the most efficient rate (779). Experienced runners in the study selected a more efficient SR as compared with novice runners. While both novice and experienced runners didn't self-select the most optimal SR, the trend demonstrates that the more experienced runners are, the closer they run to their ideal SR.

The good news is that simply by running more, runners can improve their running economy (778). While runners can get closer to their ideal running form, they will likely not self-select the most optimal form/SR.

What does this mean for you, the coach? It means that you play a valuable role in assisting your client to develop proper form and all aspects (i.e., stride rate, biomechanics, stride length, etc.) that go along with improving running economy.

FREE ENERGY

Within the body exist structures and movement patterns that allow runners to significantly reduce muscle activation while maintaining the same, if not greater, running performance. The two areas that this certification will focus on in regard to free energy are:

1. Energy Return
2. Passive Movement

Passive movement and energy return are interrelated. The genesis of free energy is that when the feet impact the ground while running, energy is absorbed and stored. The goal of a runner is to optimize running form (mechanics) to convert as much of this stored energy as possible into forces that assist in forward movement.

In regard to improving efficiency and performance, the five areas that this certification will focus on in relation to free energy is hip rotation in the transverse

plane, hip flexion, foot plantar flexion, knee flexion/extension, and foot-arch compression.

Maximizing runners' free energy contribution can improve their running economy. By utilizing free energy, active muscle requirement is reduced while still producing the required amount of force (620).

STRETCH-SHORTENING CYCLE (SSC)

“Energy cannot be created or destroyed; it can only be changed from one form to another.” – Albert Einstein

This quote pertains to a law of physics, the *conservation of energy*. In regard to the SSC, it is applicable as potential elastic energy is converted to kinetic energy.

By definition, the SSC is representative of an eccentric contraction of a muscle followed by a rapid concentric contraction of the same muscle. In respect to running, the quadriceps, obliques, and calves represent the muscles most influenced by the SSC.

When the feet hit the ground, the quadriceps and calves eccentrically contract (lengthen) and then rapidly concentrically contract (shorten) during the drive phase to provide forward propulsion (616). The stretch aspect of the muscle and tendon during the SSC is quite small, 6 to 7 percent (615).

To illustrate how the **stretch-shortening cycle** works, think of shooting a rubber band. The more you stretch the band, the farther it will travel in the air once released. However, if you stretch the band too much, it could break. Additionally, the more tension the band has (harder to pull back), the farther it will go. In respect to the human body, muscles and tendons are the rubber band. To recap, tendons connect bone to muscle, and therefore muscles and tendons are often considered two parts of a working whole, the **muscle-tendon unit** (524). The variables that affect the degree of elastic return of the SSC are (525):

1. Length of the stretch
2. Speed of the stretch (loading)
3. Stiffness of the muscle and tendon
4. Time between the stretch and the contraction

From a running perspective, the legs act as springs. The springs compress during the first half of the support phase and rebound during the drive phase.

The stiffer a muscle is, the greater the amount of energy that can be stored and released. However, to not increase the chance for injury, a muscle must have full mobility (608, 523).

Tendons

It is important to note that depending on the location of a tendon in the body, it will vary in thickness, shape, and length. These variables affect the stiffness of the tendon and thus the capacity for force production. Tendons have elastic properties that allow them to stretch. The stretch of a tendon (or muscle) stores energy, and when the stretch is unloaded, the stored energy is released. Utilizing this stored energy properly can greatly minimize the metabolic cost of movement (521).

The optimal stretch of a tendon should be viewed as a modified bell curve – meaning, too little or too much of a stretch is not optimal. If stretched beyond the end point of its range of motion, a tendon could tear completely. However, before this point is reached, a tendon could still be overstretched. When this occurs, structural changes occur to a tendon that effectively alters tendon length and thus reduces the stretch reflex (522). The degree of stretch to a tendon that elicits the ideal stretch reflex is called the **elastic region**. Once the stretch extends past this region, it is called the **plastic region**. It is at this point that the structure of the tendon changes and therefore changes the tendon length (522).

Figure 7.6 illustrates the elastic and plastic regions on a curve-based model (load – deformation curve).

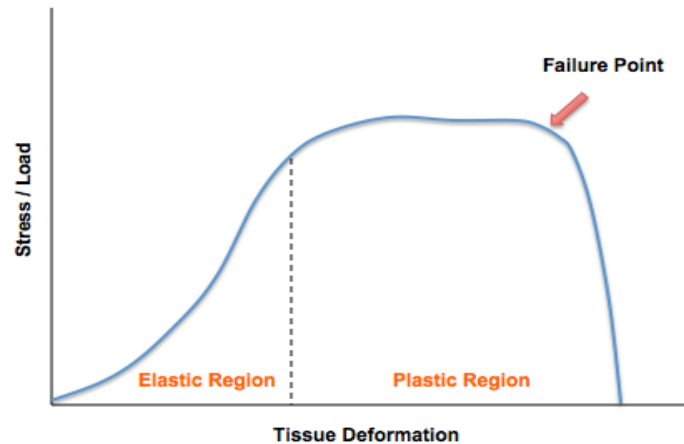


Figure 7.6 Load – Deformation Curve

Achilles Tendon

In regard to running, the Achilles tendon has a large impact on performance in respect to free energy. A study that examined the link between resistance training and Achilles tendon stiffness found that a 16 percent increase in tricep surae tendon (gastrocnemius, soleus, Achilles tendon) stiffness via resistance training decreased the rate of oxygen consumption during running by 4 percent, thereby increasing running economy (530). Another study confirmed these findings by noting that differences in Achilles tendon mechanical properties were primarily influenced by muscle strength (531).

While the amount of energy that can be stored in the Achilles tendon and the resulting increase in running economy vary, any increase in muscle strength of the tricep surae will increase Achilles tendon stiffness that results in increased running economy.

There appears to be no difference between men and women in regard to the effect of Achilles stiffness on running economy (529).

MECHANICS

As rapid dorsiflexion (loading) and plantar flexion (unloading) of the ankle is responsible for the spring action of the Achilles tendon, if the ankle is not dorsiflexed enough, the energy stored in the Achilles tendon will not be as high as it could be. The implication from a mechanical perspective is that the heel of the foot during the drive phase should touch the ground

while the hip is extended. If the heel does not touch the ground during this phase, potential energy and running economy is reduced.

Shoes with a high heel in relation to the forefoot (large heel-to-top drop) reduce the amount of stored energy. This is because of reduced ankle dorsiflexion, which places the Achilles tendon and calf muscles in a shortened position. Insufficient range of motion (flexibility) of the Achilles tendon also limits the amount of potential energy able to be stored.

OUTSIDE INFLUENCES

Running economy is not just based on storing energy but also utilizing it. The interface of the foot and the ground influence the amount of energy that can be utilized. The more solid the interface is, the more efficient a runner will be. If the running surface is very soft (e.g., sand) and/or a shoe has a lot of cushioning, this will reduce the amount of energy that can be used by the runner as a large portion of the energy is being absorbed by the ground and shoe, respectively.

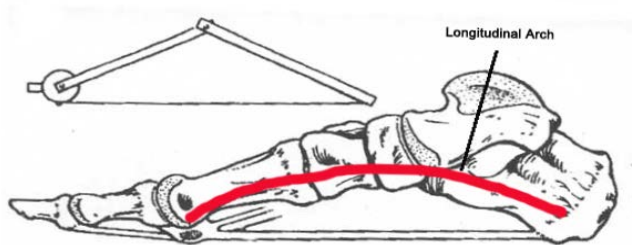
Some absorption, however, is likely a positive thing in relation to running economy and decreasing the chance for injury. Studies have shown that there is a higher metabolic cost to running barefoot than with minimal shoes because of increased muscular demand (532).

TRAINABILITY

Many distance runners do not do perform strength and sprint training because of the perceived lack of specificity. While there are many benefits to distance runners, if they did sprint training only for the purpose of stiffening the Achilles tendon to more efficiently utilize elastic energy return, it would be well worth the effort.

Foot Arch

The primary functions of the foot in regard to running are shock absorption and propulsion. The foot plays a large part in respect to free energy, specifically, the longitudinal arch (figure 7.7). The muscles and connective tissue of the foot affect the longitudinal arch, which acts as a spring to provide passive energy while running (601). Figure 7.7 denotes how the foot acts as a spring in relation to the plantar foot muscles and connective tissue.



"PF-PlantarMove". Via Wikipedia-<http://en.wikipedia.org/wiki/File:PF-PlantarMove.jpg#mediaviewer/File:PF-PlantarMove.jpg>

Figure 7.7 Longitudinal Foot Arch

A common argument for running in bare feet or in minimal footwear is that since the arch of the foot is supported in conventional running shoes, the elastic energy of the longitudinal arch is negated – thus greatly reducing an energy source used for forward propulsion.

Muscles

Like tendons, the stiffer a muscle is, the greater the elastic return will be. Preactivation of leg muscles when running prepares the body and leg musculature for foot impact. Preactivation of leg muscles is theorized to decrease stress to leg muscles and increase the cushioning upon landing (520). If the leg muscles did not contract before and during landing, the body would collapse upon foot strike.

Leg muscle “stiffness” is controlled consciously and unconsciously (526). The degree of leg stiffness directly affects the amount of knee flexion. In regard to conscious control of leg muscle stiffness, an individual is able to control stride rate and stride length (527). Leg stiffness is also influenced by the geometry of the leg at impact. This is because depending on the angle of the leg at foot impact, there will be varying loads on the leg that the muscles must counteract (526). The type of surface that the foot lands on also correlates to the degree of muscle stiffness. When running, the body looks to maintain the same degree of total vertical stiffness (surface stiffness + leg stiffness) at all times. Therefore when running on different surfaces such as pavement and sand, the degree of leg stiffness will change to maintain the same degree of total vertical stiffness. As

an example, if running on sand, the legs must become stiffer to compensate for the decrease in surface stiffness (528).

PASSIVE MOVEMENT

Go for a walk. Did you consciously contract muscles to swing your arms? This is representative of **passive movement**. Passive movement is representative of body movement that is not the result of active muscle contractions (519, 520). Specific to running, several body movements are influenced exclusively by passive movement:

- Lower leg (tibia/fibula) swing
- Hip flexion
- Knee flexion

The degree of passive movement is largely dependent on actions done prior to the passive movement occurring (520). For example, the lower leg swing (the aspect of the gait when the foot is in the air) is the result of how extended the hip is at the end of the drive phase. The greater the hip extension is, the larger the lower leg swing will likely be (think: pendulum). When examining the movement of the leg swing, the knee flexes after the foot leaves the ground behind the runner and extends prior to the foot hitting the ground. These leg movements are synonymous with hamstring and quadriceps activation, respectively. However, in respect to the leg-swing aspect of the running gait, these muscles are not activated because of passive movement.

In respect to hip flexion, during the middle of the leg-swing phase, the hip flexes to allow the leg to move forward so that the foot can strike the ground and propel the body forward. In isolation, the hip flexors are largely responsible for flexing the hip. While running, this is a passive motion, and therefore minimal muscle activation of the hip flexors occurs. One possible exception would be running uphill, in which case the hip flexors may be consciously activated.

While a primary action of the hamstrings is to flex the knee, during the gait cycle this action is largely passive versus active. Therefore the hamstrings are most utilized while running to decelerate the lower leg during the swing phase and to assist in hip extension.

Training Implications

The sport of running traditionally incorporates a lot of so-called sport-specific drills. Two of the more common drills are **butt kicks** and **high knees**. Butt kicks are exactly what they sound like – runners stand in the same spot and try to “kick their butt” with their heels. This rapid knee flexion/extension is meant to train a runner to develop a fast kick and leg turnover. Muscularly speaking, this drill primarily focuses on the hamstrings. However, as noted previously, this aspect of the running gait is largely passive. Therefore by performing this drill, the runner is taking a passive movement and turning it into an active one (e.g., muscle activation required) (579). While there may be some value to this drill from a neuromuscular standpoint, it is advised that a drill such as *strides* would be more beneficial.

The *high knees* drill is commonly performed to teach a runner how to “drive with the knees” and to focus on sprint form – as the arms are supposed to swing in relation to the legs. Like the butt kick drill, knee flexion is primarily a passive movement. This drill overemphasizes the hip flexors. Also like the butt kick drill, the high knees drill takes a passive movement and turns it into an active movement.

While the aforementioned drills appear to mimic running form, they might do more harm than good because of their lack of practicality and increased chance for injury due to their ballistic nature. There is no need to train a passive movement in isolation.

❖ ALLOCATION OF ENERGY

Watch an individual run. There is a lot going on – legs and arms swinging, torso rotating, ankles flexing and extending. This leads to the question, how much energy is required to make each of these individual things occur and how does it relate to an individual’s total energy expenditure? Fortunately, Dr. Rodger Kram from the University of Colorado wondered this as well and performed several studies to determine the answer (584–588).

Dr. Kram et al. discovered that the following areas were responsible for consuming the most energy while running (585):

- 69 percent – Supporting body weight
- 29 percent – Forward propulsion

Taking muscle synergy into account, Kram et al. deduced that supporting one's body weight and forward propulsion equates to 98 percent of the energy used by a runner (587). While these percentages can likely be varied to some degree based on the efficiency of a runner, the primary thing to take note of is the distribution of percentages as it relates to areas of focus. For example, it would be most advantageous from a performance standpoint to reduce body weight.

As noted previously, Dr. Kram et al. found that swinging the arms reduced a runner's energy cost by approximately 4 percent (586). This is likely due to the counterbalancing effect.

Lastly, Dr. Kram et al. found that the balance required by a runner equated to 2 percent of their energy cost. The remaining aspects that likely influence energy cost of a runner are the running surface, shoes, and step width (587, 588).

❖ ROTATIONAL-DRIVEN RUNNING

While hip (pelvic) rotation in the transverse plane is an important aspect of the running gait, it is also one of the least-discussed aspects of running form. By leaving out pelvic rotation, a major aspect of force production is eliminated.

The rotational concept of locomotion was introduced by Dr. Robert Lovett in the early 1900s (621), developed upon by Dr. Serge Gracovetsky (617), and modified further by Dr. Erik Dalton (619). Prior to the introduction of this concept of locomotion, the traditional locomotion model was called the *pedestrian model of gait*. This model states that the legs are the primary aspect of the body responsible for locomotion and the upper body is largely passive (622).

Rotational-driven running reduces compression on the spine by adding a rotary element. This rotation stores and releases energy via torsion (775).

THE SPINAL ENGINE

This model (*spinal engine*) was developed by Dr. Gracovetsky. The catalyst for the spinal engine originated by observing an individual with no legs walking upright by rotating the pelvis, thereby “walking” on the ischial tuberosities (landmarks on the base of the pelvis) (617). This model consists of two primary aspects (617):

1. Spinal Rotation
2. Coupled Motion

A key aspect of the *spinal engine* model is that unlike the *pedestrian model of gait*, Dr. Gracovetsky theorized that upon foot strike while running, energy was not absorbed by the ground but rather utilized up the body’s kinetic chain to assist with locomotion (619).

As referred to earlier, this relates to *ground reaction force*, which is based on Newton’s third law:

For every action, there is an equal and opposite reaction.

Here is how the *pedestrian model of gait* and the *spinal engine* interpret ground reaction force (Newton’s third law).

- **Pedestrian Model of Gait:** Energy is absorbed by the body.
- **Spinal Energy:** Energy is utilized via counter-rotation of the spine to facilitate locomotion.

Spinal Rotation

When running, the spine rotates (i.e., twists) in opposite directions. This is representative of counter-rotation of the hips and torso, as noted in figures 7.8 and 7.9. This counter-rotation involves the *stretch-shortening cycle* (SSC) to utilize potential energy for the purposes of stabilizing the body and assisting in locomotion via transverse rotation of the hips (774).

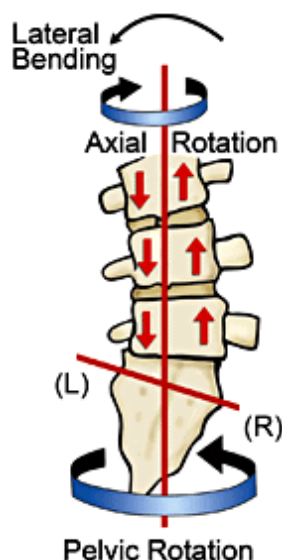


Figure 7.8 Spinal Counter-Rotation

Used with permission from Dr. Erik Dalton

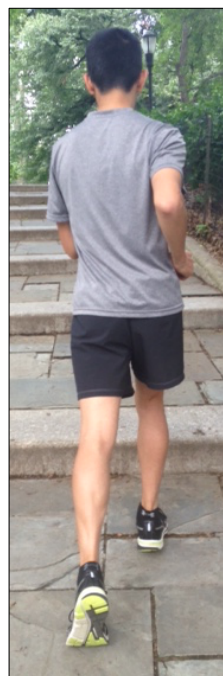


Figure 7.9 Spinal Counter-Rotation

Figure 7.8 illustrates how the spine counter-rotates and laterally flexes while running. Spinal counter-rotation is represented in figure 7.9. In this image, you can see that the runner's shoulders (axial rotation) are rotated to the right while the pelvis is rotated to the left as the left hip is extended. Figure 7.9 is a real-life example of figure 7.8 (note the slight lateral spinal flexion to the left).

It is normal for individuals to differ in regard to how much spinal counter-rotation they have. A rough assessment for this is to have clients lie on the floor with their knees and hips at 90 degrees and have them rotate their lower body down to one side without raising their shoulders off the ground. Some individuals can rotate their legs all the way down to the ground whereas others may be able to get their legs only halfway down before the contralateral shoulder lifts off the ground.

Regarding movement in the transverse plane, the shoulders rotate much more than the hips (pelvis). As a result, it can be difficult to decipher visually if an individual is running with transverse hip rotation. Video 7.3 denotes transverse rotation of the hips as it pertains to hip extension and counter-rotation of the spine.



Video 7.3 Hips with and without transverse rotation

SPIRAL SPRING SYSTEM

According to Dr. Gracovetsky, intervertebral disks of the spine are not for the purpose of shock absorption but rather to store and unload energy via torsion of the spine (775).

The best way to think of Dr. Gracovetsky's spinal engine is of the spine as a long spring. Visualize a long metal coil spring – using both hands, grab the coil spring at each end. Twist the spring in opposing directions and then let it go. The coil will allow some rotation to occur, and when you let it go, it springs back to the starting position. The twisting of the spring in opposing directions is representative of spinal rotation and thus transverse rotation of the hips and torso.

While the spinal engine model states that most of the energy stored via the spiral spring system is located in the collagen of the vertebral disks (annulus) and the fascia of the low back, other aspects of the body, such as the core musculature (inner-/outer-unit core muscles), play a role in the storage and release of energy (775).

The muscles involved in the spiral spring system are noted in the next section, Coupled Motion.

As it pertains to running, the spinal engine (SE) is associated with a heel-strike gait and, more specifically, the stretch of the biceps femoris during a heel strike. As a result, running with a midfoot or forefoot strike may reduce the activation of the biceps femoris in regard to being a major contributor to the spinal rotation. This is described in greater detail in the PSSS section.

Coupled Motion

With proper running form, power is initiated through rotation of the hips and the resulting follow-through of the legs. This principle is the same as the boxer described earlier regarding *kinetic linking*. In that example, for the boxer to effectively throw a punch, the following things occurred sequentially:

1. Hips rotate
2. Rotation of the torso
3. Arm extension

This allows the boxer to throw a powerful punch with the origin of the punch coming from transverse hip rotation.

Dr. Dalton modified Dr. Gracovetsky's spinal engine model (myoskeletal engine model) to include a modified version of the body's natural spring system (774). In doing so, Dr. Dalton identified what he termed the **anterior spiraling spring system (ASSS)** and the **posterior spiraling spring system (PSSS)** (776). These spring systems comprise muscle and fascia (myofascial). These spring systems, in combination with spinal rotation, represent the "engine" that powers proper running locomotion. In other words, these spring systems work together to facilitate proper running form.

Dr. Dalton identified two other spring systems that aid in locomotion, the **lateral spring system** and the **stirrup spring system**. However, this certification is focused on the ASSS and PSSS systems, as they represent the most relevant systems in regard to spinal rotation.

As an example, while the spine bends laterally slightly during the gait cycle, this lateral bending (lateral spring system) is more of an automatic process whereas running with proper spinal rotation is not.

HOW DO THE SPRING SYSTEMS WORK?

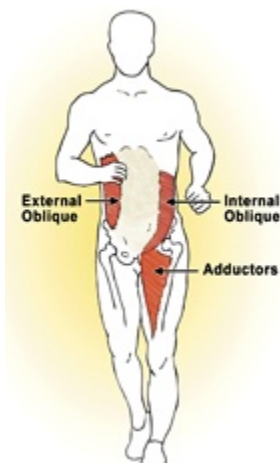
The hips (pelvis) and shoulders rotate in opposite directions.

1. When the hips and shoulders are counter-rotated the most, this represents the point at which the muscles, tendons, and fascia of the spring systems are stretched the most. This relates to the point in the gait cycle where the most potential energy is stored. This also corresponds to the legs being the farthest apart (fore/aft) in the sagittal plane.
2. As the hips and shoulders begin to rotate back toward each other, this is representative of the potential energy stored in the spine, muscles, tendons, and fascia converting to kinetic energy to rotate the spine and thus the hips and shoulders.

3. When the hips and shoulders are in the same frontal plane (the legs pass by each other under the torso at this point), this represents the least tension (stretch) on the spring systems and also represents the point at which the stretch of the spring systems alternates to the opposing side of the body.

ANTERIOR AND POSTERIOR SPRING SYSTEMS

The **anterior spiraling spring system** comprises the following muscle groups (776):

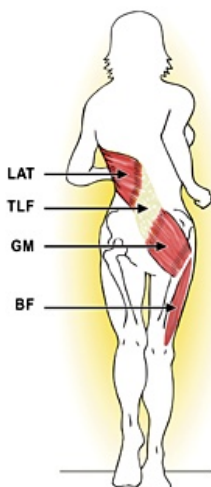


- External Oblique
- Abdominal fascia and external oblique aponeurosis (tendon sheath)
- Contralateral Internal Oblique
- Contralateral Adductors

Used with permission from Dr. Erik Dalton

Figure 7.10 ASSS

The **posterior spiraling spring system** comprises the following muscle groups (776):



- Latissimus Dorsi
- Fascia of the thoracic and lumbar spine region
- Contralateral Glute Max
- Contralateral Biceps Femoris

Used with permission from Dr. Erik Dalton

Figure 7.11 PSSS

These spiraling spring systems (PSSS and ASSS) gain and release energy via the stretch-shortening cycle. As you can see from the preceding PSSS and ASSS images, the spring systems form diagonal lines. These diagonal lines stretch and contract in a kinetic chain to facilitate spinal rotation and therefore transverse pelvic rotation. Stretching and contracting of the PSSS and ASSS are representative of an elastic response.

The PSSS and ASSS oppose each other, meaning the ASSS crosses the body diagonally in the front whereas the PSSS crosses the body diagonally in the back, and at an opposing angle (figure 7.12).

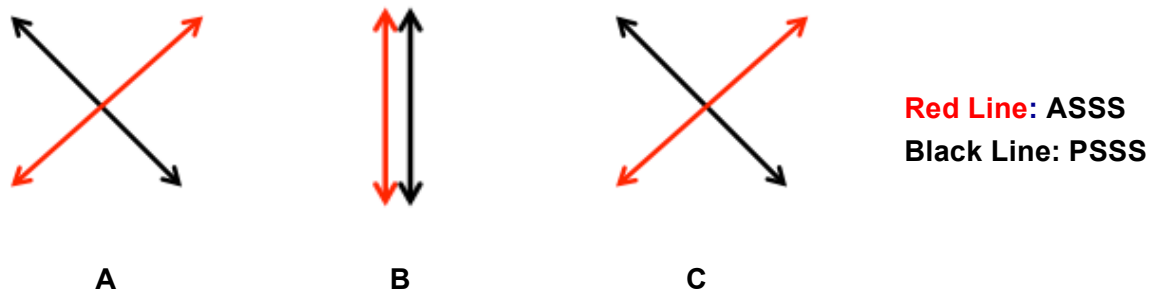


Figure 7.12 PSSS/ASSS Angles

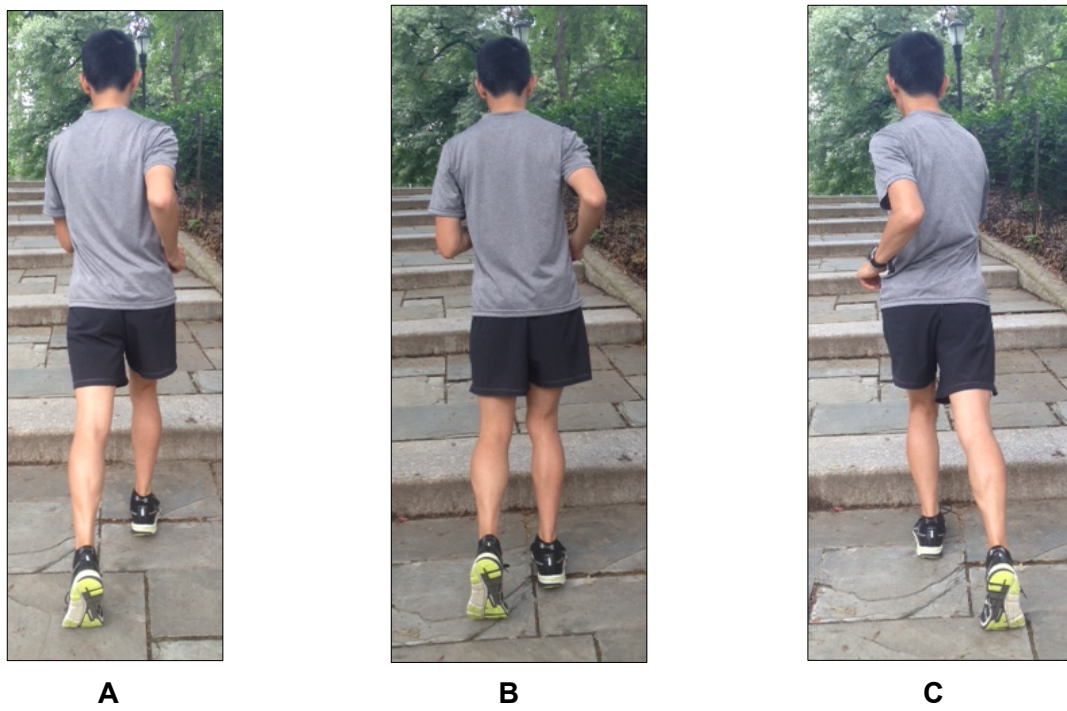


Figure 7.13 PSSS/ASSS Gait

- A:** Right leg forward, left leg rearward, pelvis rotated left, shoulders rotated right
- B:** Both legs directly underneath pelvis, pelvis and shoulders aligned in frontal plane (no stretch of either ASSS or PSSS system)
- C:** Left leg forward, right leg rearward, pelvis rotated right, shoulders rotated left

In addition to Dr. Gracovetsky, Dr. Dalton's PSSS and ASSS systems were influenced by Tom Myers (Anatomy Trains) and Dr. Andry Vleeming, who developed his own muscular spring models: anterior oblique system and posterior oblique system (777).

ARM MOTION

As the latissimus dorsi is primarily stretched by forward movement of the arm (humerus), running with no arm movement takes away an element of the ASSS and therefore reduces the total force of the spiral spring system. Conversely, too much arm motion wastes energy. Like most elements of human performance, the optimal amount of arm swing/movement exists on a bell curve.

Did you ever wonder why track sprinters have very muscular upper bodies and a lot of arm movement compared with distance runners?

Did you ever notice that when you go from running at a slow pace to a sprint, your arm movement becomes more pronounced?

This is because when more speed is required, the body needs to generate more energy. This is accomplished by increasing spinal rotation, and therefore a larger and more rapid stretch/contraction of the PSSS and ASSS is elicited. By increasing arm movement, a greater stretch is placed on the latissimus dorsi, thus increasing the potential and kinetic energy of the PSSS.



Video 7.4 Arm movement

'THE ARMS DRIVE THE LEGS'

This is a common explanation as to why arm movement while running is important. In the simplest of terms, arm motion does not drive the legs. As discussed previously, arm movement is primarily used to assist in counterbalancing the body and stretching the latissimus dorsi (via PSSS) to create potential energy to assist in spinal rotation.

Therefore arm movement can assist in driving the legs (via spinal rotation). However, it is likely that this is not what most people mean when they refer to this statement.

RANGE OF MOTION

Too much or too little transverse spinal rotation will likely decrease the effectiveness of the spinal engine. Too little rotation will not create enough stored (potential) energy whereas too much rotation will decrease running economy because of excessive motion. The ideal amount of rotation exists somewhere in between the two extremes.

SUMMARY

Specifics of the SSC and spinal engine can get quite complex, especially in regard to the exact muscles and aspects involved with spinal rotation. However, the following sums up spinal rotation in regard to its actions and benefits:

Counter-rotation of the spine facilitates hip extension via hip rotation in the transverse plane. This allows for increased glute and core (primarily obliques) involvement and reduced ground reaction force (GRF). The result of this is less reliance on the hamstrings, reduced compression of the spine, and increased force production.

Net effect = A faster and more efficient/economical runner due to a reduced metabolic cost of gait.

Relationship between Hip Rotation and Running Speed

Akin to the relationship between foot-strike location and speed, the amount of hip rotation is typically proportional to the speed of a runner. While there is a maximum degree of hip rotation that is considered optimal regardless of speed, the greatest amount of transverse hip rotation is often seen in middle-distance runners (400–1,600 meters) and sprinters.

As we know, the latissimus dorsi is part of the spinal engine kinetic chain (PSSS) and therefore an important aspect of spinal counter-rotation. Take a look at a 100-meter sprinter compared with an elite marathon runner – quite different body types. The most noticeable difference is the size of the upper body. If the *pedestrian model of gait* were correct in that the upper body in regard to running is just along for the ride, then a muscularly developed upper body would be dead weight and slow the runner. This, of course, is not true. The most likely reasons for a sprinter's developed upper body are:

- Substantial arm movement in the sagittal plane
- Large degree of spinal counter-rotation (hip rotation)
- Because of the high degree of spinal torsion and fast leg turnover, substantial strength is required to maintain proper form and core stabilization.

Conversely, let's examine the form of someone walking or slowly jogging. While there is some degree of spinal counter-rotation, the slow speed does not require a substantial amount of it. As a result, vertical arm movement, transverse hip rotation, and leg/hip extension are on a much smaller scale as compared with that of a middle-distance runner and certainly that of a sprinter.

So what does this mean? At slow speeds (i.e., walking, slow jogging), there is minimal transverse hip rotation, which therefore increases the reliance on leg muscles (especially hamstrings) for hip extension and overall locomotion. At faster speeds, transverse hip rotation via the spinal engine becomes more important for locomotion. Individuals who run at higher rates of speed without adequate transverse hip rotation likely overstress the hamstring musculature, hyperextend the lumbar spine, and overstretch the hip flexor musculature.

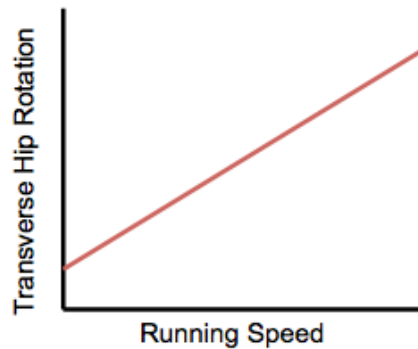


Figure 7.14 Correlation of speed/hip rotation

To test the validity of figure 7.14, start off walking and take notice of your arm movement and transverse hip rotation. Now, begin to jog slowly – then increase your speed. You will notice that as you increase your speed, your arm motion and transverse hip rotation increase. While increased arm movement is natural at higher speeds, some individuals compensate for a lack of transverse hip rotation with a faster leg turnover. While a fast leg turnover (cadence/stride rate) is required to run faster, it should occur in tandem with an increase in transverse hip rotation.

Relationship between Spinal Rotation and Efficiency

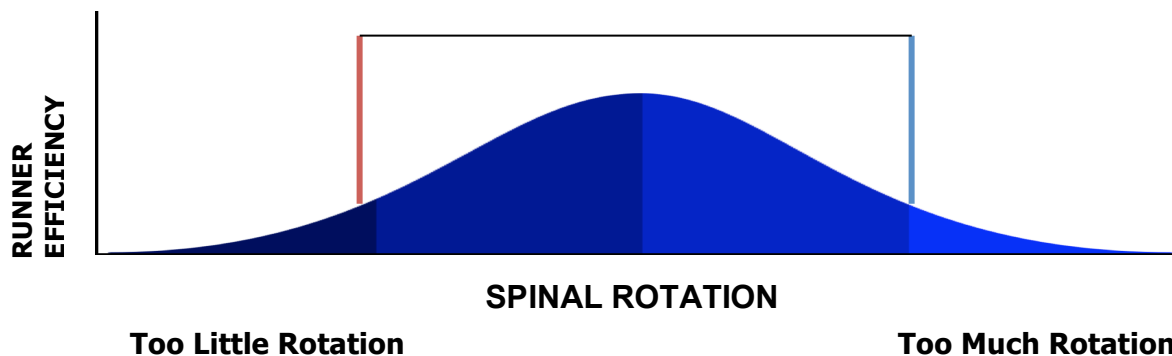


Figure 7.15 Correlation of spinal rotation/efficiency

The speed of the runner affects the optimal degree of spinal rotation. In relation to the total spectrum of hip rotation, it is likely that a slow running speed would relate to a small amount of spinal rotation being most efficient whereas a fast running speed would likely translate to a greater amount of spinal rotation being optimal.

In figure 7.15, the rectangle denotes the optimal degree of spinal rotation, with the vertical red line denoting ideal spinal rotation when running slowly and the blue vertical line denoting ideal spinal rotation when running fast. Therefore the degree of spinal rotation outside the rectangle denotes too little and too much spinal rotation. Be aware that *slow* and *fast* are subjective based on the individual.

In the following video, focus on the yellow marker placed on the greater trochanter (top of femur). As the video progresses from minimal transverse hip rotation to substantial hip rotation, notice the increased horizontal movement of the marker.



Video 7.5 Hip rotation

While transverse hip rotation should occur at all running speeds, it often does not.

Elastic Response

As noted previously, the elastic stretch response is part of the stretch-shortening cycle (SSC). As the hips rotate in the transverse plane in one direction, the torso rotates in the opposing direction. This stretch involves the intervertebral disks and muscles/tendons/fascia noted in the preceding PSSS and ASSS illustrations (774).

Increased muscle tension and stiffness increases elastic response. Therefore the more engaged the core musculature is, the greater the elastic return of the hips will be.

Hip rotation (toward the side of the drive leg) begins before the foot hits the ground. This illustrates how hip rotation initiates the running drive phase and how the core musculature plays a key role in force production while running.

Cueing Hip Rotation

For runners who do not rotate or over-rotate the hips when running, learning the correct amount of transverse hip rotation can be hard. The degree of hip rotation

is directly correlated to the degree of rotation of the torso and, more specifically, the shoulders. Don't believe it? Try running without rotating the upper body while rotating the hips ... awkward! Conversely, try running while substantially rotating the torso but not rotating the hips. As you can see, this does not work very well. The mechanical reason for this is counterbalance. As defined by *Merriam-Webster*, counterbalance is (533):

A force or influence that offsets or checks an opposing force

Be aware that when teaching clients to run with the proper mechanics and, more specifically, the correct degree of hip rotation, you should not expect to see them “get it” immediately. Like other areas of motor development, it is a process and individuals will acquire proper form at different rates.

The most important aspect when teaching proper form is that clients fundamentally understand why the correct technique is more efficient and economical than what they are currently doing and understand the theory and mechanics involved. Even once a client is able to run with proper form, it takes a lot of practice for it to become repeatable to the point where it becomes the normal, default form.

Cue Shoulders

Because of spinal counter-rotation, if your client under or over-rotates the hips, a good starting place is the shoulders. By instructing your client to increase or decrease rotation of the shoulders, it is likely you can directly affect the rotation of the hips. To demonstrate this, clasp your hands together with your arms pointed straight ahead. Now begin running and make sure there is no lateral movement of the arms. This prevents the torso (shoulders) from rotating, which in turn greatly reduces hip rotation because of a lack of spinal counter-rotation.

RUN FROM THE SHOULDERS, NOT THE ARMS

While countless resources discuss “correct” arm position and movement, there are few, if any, resources that discuss proper torso rotation. As a result, most runners focus on arm movement and not torso rotation. As the following image (figure 8.9) denotes, an effective way to cue runners that under-rotate their torso is to have them focus on moving their shoulders forward in the sagittal plane while running.

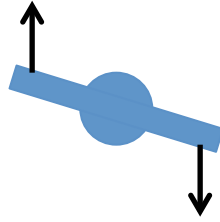


Figure 7.16 Cueing shoulder rotation

This “cue” (fig. 7.16) assists runners in attaining a greater degree of torso rotation. The purpose of this drill is not to have clients attain the perfect degree of torso rotation, but rather to have them understand how shoulder movement affects torso rotation and to have them “feel” how increased torso rotation affects hip rotation and gait mechanics. Once clients have greater awareness of these areas, they can work to adjust their mechanics to achieve the optimal degree of shoulder/torso rotation.

Cue Leg/Hip Extension

Another way to cue hip rotation is to instruct runners to focus on getting full leg/hip extension behind their body. Cueing clients to push down and back with a foot until the leg is at full extension will cause the hips to rotate to the side of the extended leg. While this may overexaggerate hip rotation initially, it can be a valuable learning tool for clients so they understand the link between leg extension and hip rotation.

Cue Arms

While the arms in isolation do not aid in locomotion, fore/aft arm (humerus) movement contributes to stretching the latissimus dorsi, which is part of the kinetic chain involved with the posterior spiraling spring system (PSSS).

Correlation between Ankle Flexibility and Hip Extension



Figure 7.17 Calf Stretch

Hip extension is typically related to transverse hip rotation, hip flexor flexibility, and glute max/hamstring activation. This is true from a static position. However, in regard to running and, more specifically, proper running form, when the leg is in the drive phase (posterior to the body), the ankle must have adequate dorsiflexion or else the drive phase of the stride will be cut short – thus reducing hip extension.

When people discuss *shortening one's stride*, this is (or at least should be) referring to reducing stride length in front of the body, not behind it. While reducing the stride length in front of the body (termed *overstriding*) is typically a good thing, the posterior stride

length should not be shortened. In fact, for most runners, the posterior aspect of the drive phase is too short.

Therefore if your client has poor ankle flexibility (i.e., poor ankle dorsiflexion), you should work on stretching the calf muscles (gastrocnemius and soleus) as well as utilizing a foam roller. When stretching the calf, ideally it will be in a position that is specific to running – hip extended with the leg straight and the foot as flat to the floor as possible and without pain.

If you have clients who run with a very short posterior stride and/or prematurely lift their heel during their posterior stride (or always run on their forefoot), a lack of adequate ankle dorsiflexion may be the culprit.

❖ MIDFOOT STRIKE

In recent years, the number of runners utilizing and converting to a midfoot strike has increased dramatically. This has coincided with an increase in minimalist running shoe usage. A catalyst for the midfoot/minimalist running movement is the book *Born to Run* by Christopher McDougall (69). This book profiles the Tarahumara tribe in Mexico, in which some members run long distances wearing

little more than thin rubber soles. The book not only discusses footwear but also the midfoot-strike running style.

As noted previously, an important factor in an efficient running gait is the foot striking the ground below or slightly in front of the body. Many runners find that this form is best facilitated via a midfoot strike.

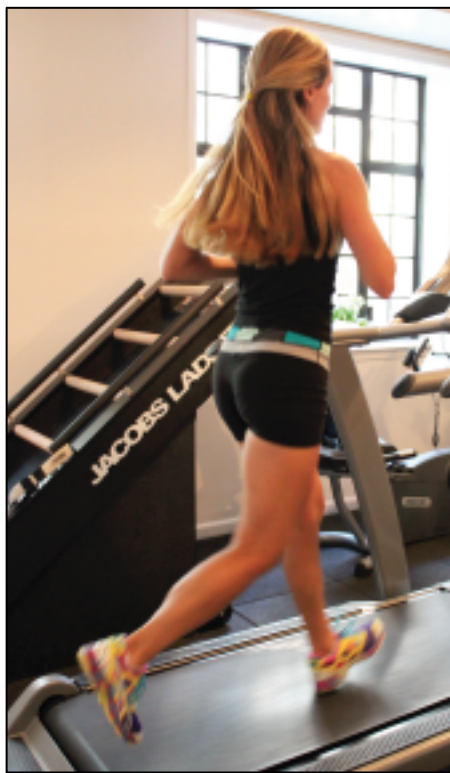
Is a Midfoot Strike Right for Everybody?

Probably not. As is noted elsewhere in the certification, a heel strike puts more stress on the skeletal system whereas a midfoot strike puts more stress on the muscular system – especially the calves. Some clients, regardless of the length of adaptation, will have muscular/connective tissue issues when running with a midfoot strike.

This section is included to discuss the pros and cons of a midfoot strike, not to necessarily encourage you to convert your clients to this foot-strike type. This must be ascertained on an individual basis. Like most other areas of training, some clients will find more success with one type of training (or foot-strike type) than others.

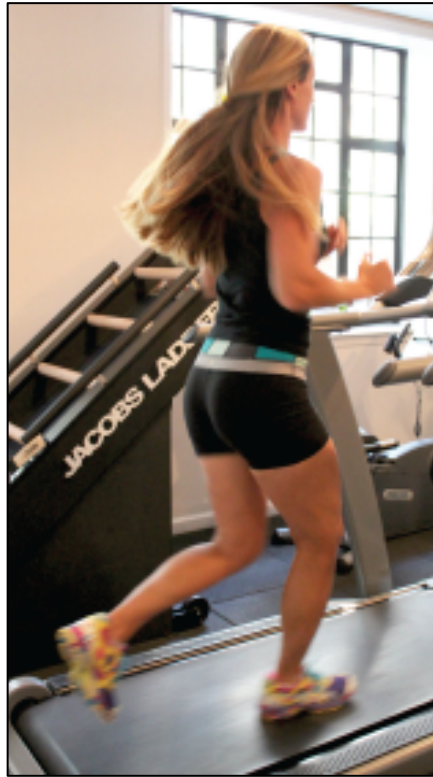
For many runners, the topic of foot strike is quite polarizing. This section examines the differences between foot-strike types and the research that surrounds this topic.

MIDFOOT STRIKE vs. HEEL STRIKE



Heel Strike

VS.



Midfoot Strike

Figure 7.18 Heel strike vs. midfoot strike

When comparing these two running styles, biomechanically, a midfoot strike produces less impact force on the body than running with a heel strike (321). This is because a heel strike is more reliant on the skeletal system for support and stability than a midfoot strike, which relies more on the musculature system.

If your client heel strikes, the individual should not necessarily convert to a midfoot strike.

As a midfoot strike tends to reduce the impact on the knees, hips, and back, and lessen stress on the hamstrings, if your client is not experiencing any issues in these areas, the *if it ain't broke don't fix it* theory may be applied. However, if you feel that because of biomechanical or flexibility reasons your client might benefit from a midfoot strike, it would be advantageous to speak with the person about slowly converting. A midfoot strike tends to put more stress on the soleus and Achilles tendon, especially for those converting to this foot-strike type.



Video 7.6 Differing foot strikes (heel, midfoot)

- Notice the difference in the rate of leg turnover between the two foot strikes

While a midfoot strike reduces the magnitude of impact forces on the feet relative to a heel strike, it does not mean that it will feel better or be more comfortable, especially to those in the process of converting (321). You must be patient with your clients while they are making the switch from heel to midfoot running. Remember, your clients have been walking and running with a heel strike their whole lives, so change will most likely not come quickly, easily, or without some degree of frustration.

As noted previously, a heel strike that occurs substantially in front of the body (overstriding) puts the hamstrings under substantial eccentric stress. If the stride is really long, hyperextension of the knee is also possible. When running with a midfoot strike, the aforementioned issues are largely corrected. This results in the ability to run with less chance of injury to the hamstrings.

It is important to discuss several misnomers in relation to running stride.

Running with a Heel Strike = Running with the Brakes On

This is a common comparison when analyzing a heel strike in relation to a midfoot strike. This insinuates that when the heel hits the ground, it slows the runner since the leg is extended in front of the body. This is technically incorrect, as the forward leg does not act as a brake under normal running conditions. As soon as the heel hits the ground, the knee flexes, the forefoot hits the ground, and the body moves forward and over the foot of the outstretched leg. While not an overly efficient running stride from a biomechanical standpoint, the *braking* aspect of the heel strike can be more accurately described as *coasting*. This is because during the support phase, the body must “catch up” to the forward foot before the drive phase can commence. During the support phase, there is no forward propulsion generated by the lead leg. Rather, the body is coasting and using momentum to continue forward movement until the leg is under the body and can begin the drive phase.

While the body is coasting until the leg is under the body, the relative amount of time coasting is very small, especially when running at a fast pace. Therefore,

unless the stride length is extremely long and the overall speed of the runner is slow, the negative effect on a runner's speed due to *coasting* is slim to none.

To demonstrate what *braking* is, have your clients run to a set point then have them rapidly decelerate and stop. They will notice that their quadriceps and abdominal muscles have to contract to decelerate. With a normal running gait, once the foot strikes the ground, the torso continues moving forward and “over” the support/drive phase leg. When braking, the quadriceps and abdominals contract to restrict the forward motion of the torso, thus slowing the runner. As you can see, this does not occur with a normal running gait.

Ground Contact during a Midfoot Strike: Balls of the Feet Only?

Many people who adopt a midfoot strike impact the ground only with the balls of their feet, leaving the heel of the foot in the air. This places excessive stress on the Achilles tendon and calf muscles, primarily the soleus. While the emphasis of foot contact with a midfoot strike is the ball of the foot, it should not be the only point of contact. Once the ball of the foot impacts the ground, immediately thereafter the rest of the foot should lightly impact the ground. While the ball of the foot is the area that pushes off the ground, contact (ground) between the middle of the foot and heel is used for support. This support decreases the stress on the Achilles tendon and calf muscles. When watching a runner with a proper midfoot strike, it should almost look as though the entire length of the foot hits the ground at the same time. The only time when the ball of the foot should hit the ground exclusively is when running up a steep hill or sprinting. This is called a *forefoot strike*.

DRILLS: MIDFOOT FORM

To understand what a midfoot strike is supposed to look like and, more important, how it is supposed to feel, there are two very good drills:

Uphill Running

Have a client run up a gradual (2–5 percent grade) hill. The postural traits of a midfoot strike are almost identical to that of running uphill.

- Slight forward upper-body lean (from the ankles, not the hips)
- Foot lands on midfoot versus heel
- Feet land under body

Resistance Running

Another way to teach proper midfoot strike form is to wrap a long theraband or rope around your client's waist while you hold the ends behind the person. Cue the client to lean forward at the ankles, not the waist. You should be at least three feet behind your client. Then instruct the client to run forward against the resistance. During this time, you need to be braking to slow the runner. Your form should be similar to walking down a steep hill – low center of gravity, hips shifted down and posteriorly, knees bent, and upright torso. This drill is effective because the runner must lean forward and strike with the midfoot. This drill can also be accomplished by having your client pull a weight-loaded sled attached at the waist by a harness/rope.

CONVERTING TO MIDFOOT FORM

Lots of runners who convert to a midfoot strike from a heel strike will notice an increase in calf tightness and, occasionally, injury to the soleus and Achilles tendon. While soleus tightness is to be expected to some degree, injury should not be a part of this stride transition. To reduce the chance of injury and to facilitate an effective transition to running with a midfoot stride, following are several important things to address:

1. Allow the midfoot and heel to impact the ground almost simultaneously (ball of the foot impacts the ground slightly before the heel) to reduce the stress on the calf muscles.
2. Perform exercises to strengthen the soleus and Achilles tendon (e.g., jumping rope, walking on balls of feet)
3. Right before the foot hits the ground, it should be moving rearward and the hips should rotate toward the side of the body of the foot strike.

4. Do not bounce excessively when running. Bouncing causes the calf muscles and Achilles tendon to work excessively to accelerate and decelerate the body vertically. Aside from being potentially more self-injurious, it is also a slower way to run as energy should be used for forward, not vertical propulsion.
5. Gradually introduce midfoot striking into runs while wearing nonminimalist shoes. If you have clients who use motion-control or stability shoes and/or orthotics, continue to have them run in these shoes as they intersperse midfoot striking into their runs. As the client adapts to a midfoot strike, the time spent running with a midfoot strike should increase and shoe selection can gradually become more minimal, if desired. A good starting time frame is 15 seconds to a minute for midfoot running.

Athletes who convert to barefoot or minimalist running with no progression are at an increased chance for injury. Individuals converting from a heel to a midfoot strike often do not have the structural arch support or muscle and connective tissue strength to be able to run in an unsupported shoe. Muscles and connective tissue adapt to what they have been doing for long periods. Adapting to a new running shoe type or new foot strike requires proper progression.

When Not to Make Changes

If you acquire a client who is currently heel striking and has a race in the near future (anything less than five months), it is advisable to not convert the individual to a midfoot strike for the event. Running efficiently with a heel strike is a learned skill, as is running with a midfoot strike. Unless the client is becoming injured because of heel striking, trying to unlearn heel striking and learn to run with a midfoot strike when a race is not too far away is not a smart idea and will most likely lead to slower times, frustration, and possible injury. This is not to say that you cannot introduce small doses of midfoot striking into the training program, but you should not expect a client to successfully run the event with the new foot strike. All changes that are biomechanically based need to be introduced in very small doses and over extended periods.

Go Slower to Go Faster

When converting to a midfoot strike, initially the speed of a runner will likely decrease. This is normal and must be conveyed to a runner prior to and during the conversion to midfoot striking.

SHOULD I CONVERT MY CLIENT TO A MIDFOOT STRIKE?

Depends. Do you think the client could be more efficient? Is the individual experiencing issues that may be rectified with a midfoot strike? If the answer to one or both of these questions is yes, then conversion to a midfoot strike may be the right course of action.

Contrary to many running articles, striking with the heel versus the midfoot is not bad or incorrect, just different. Therefore if your client is heel striking and doing fine with that, there is no reason to change. In the same respect, if your client is currently running with a midfoot strike and is having issues (i.e., chronically tight soleus), the right course of action may be to work with your client to convert to a heel strike.

More important than the type of foot strike is the angle of the tibia upon foot strike. This is discussed in the next section.

❖ IT'S ALL ABOUT THE TIBIA

Running-gait (stride) efficiency correlates directly to the angle of the tibia in relation to the ground at the time of foot strike. Upon foot strike, the angle of the tibia in relation to flat ground should be at or close to 90 degrees.



Figure 7.19 Tibia angle: Foot in front of body

In image 7.19, the blue line represents 90 degrees and the distance between the yellow and green lines represents the acceptable tibia angle range during the foot strike. Therefore, the foot-strike angle (red line) in figure 7.19 is outside what should be considered an acceptable range.

A foot-strike (tibia) angle at or near 90 degrees in respect to the ground allows forward propulsion to occur almost immediately as the foot is directly under, or close to being under, the body's center of gravity. If the foot strikes substantially in front of the body, the ground reaction force creates an upward and rearward moment that does not facilitate forward motion (215). In other words, when the foot is too far in front of the body at the time of foot strike, a "dead spot" in the stride is created because of a lack of forward propulsion occurring with the drive leg. Research has demonstrated that landing on the heel significantly ahead of the body increases the loading rate versus a midfoot strike with the leg under or just in front of the body. It is likely that the closer a heel strike is to the body, the lower the loading rate is in relation to a heel strike farther in front of the body.

If you have clients running with a heel strike, the goal should be to shorten their stride in front of their body so the feet land under, or close to being under, the

body's center of gravity. This leads to a reduction in stride length and an increase in their stride rate (leg turnover). Additionally, a reduction in stride length reduces the angle that the foot is pointed diagonally upward when the heel impacts the ground. When the foot is pointed upward excessively, it causes the foot to “slap” down on the ground, increasing stress on the legs and body. This slap also causes a slight delay in time before the foot can act to push down and rearward to propel the body forward, though the negative effect of this on performance is negligible.

Drill

To demonstrate why the tibia needs to be at or close to a vertical position upon foot strike, have a client stand on one leg with the knee behind the ankle (the client will need to hold on to something for stabilization). Then instruct the client to push off the foot to move the body forward. The client will find that until the knee is slightly ahead of the ankle, forward propulsion is not possible. This directly correlates with the body being over the foot to generate forward propulsion.

NATURAL FOOT GAIT

Generally speaking, the faster or more uphill an individual runs, the farther forward on the foot they tend to run. Conversely, the slower a person runs (or walks), the more natural heel striking often feels. If you look at the sole of a track sprinter's shoe you will see that the spikes are under only the balls of the feet and the toes. This is because when people sprint, they are forced to emphasize forward propulsion at all times. Conversely, ask your client (or try it yourself) to walk with a midfoot strike. This is extremely awkward and feels very odd. The faster a person runs, the more natural a midfoot strike will likely feel. This is substantiated in the study discussed next.

Bare Feet = Midfoot Strike?

There is an assumption by many that the body automatically converts to a midfoot strike when running in bare feet or in minimalist shoes. The thought process behind this is that barefoot runners naturally run with a midfoot strike to avoid the high stress (impact forces) that is associated with running with a heel strike. A 2013 study out of George Washington University (GWU) found this not

to be the case (321). The idea that running in bare feet naturally converts a runner to a midfoot strike was the result of studying just one population of barefoot runners. This research was then marketed in popular running journals.

The GWU study researched 38 runners from a habitually barefoot population in northwest Kenya called the Daasanach. The study had the runners run around a track at varying speeds. The study found that while running with a midfoot strike lowered impact forces, most runners ran with a heel strike. While some runners switched to a midfoot strike while running at faster speeds, a midfoot strike was not a constant theme when running at faster speeds. This research contradicts the theory that running in bare feet automatically changes one's running gait from a heel to a midfoot or forefoot strike (321).

Foot-Contact Time

Foot-contact time relates to how long a foot is in contact with the ground during the running-gait cycle. Running with a heel strike results in a longer contact time than a midfoot strike because of the feet typically hitting the ground in front of the body, thus elongating the stride length. A 2007 study by Hasegawa et al. used video cameras at the 15K point in a half marathon to capture the foot-strike type and foot-contact time in which beginners to elite competitors competed. The results showed the following (322):

- A higher percentage of faster runners ran with a midfoot strike than slower runners
- Faster runners had less foot contact time than slower runners

While not all “fast” runners use a midfoot strike, the tendency is that the faster one runs, contact occurs farther forward on the foot. The following chart illustrates natural running gait transitions based on the running speed.

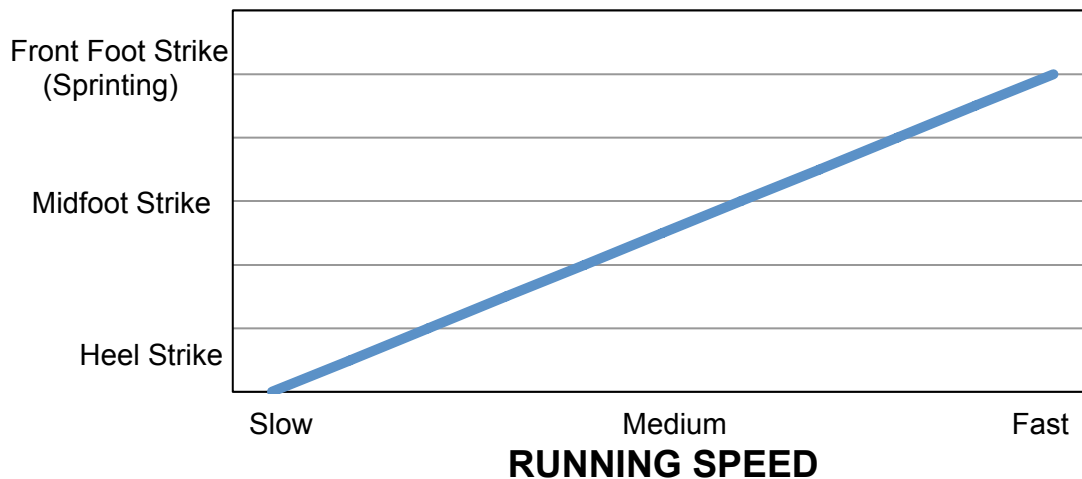


Figure 7.20 Influence of running speed on foot-strike type

Another tendency when converting to a midfoot strike or when getting tired during a run is that the foot impacts the ground while the leg is still moving forward and not rearward. This causes the stride to look like a shuffle and increases foot-contact time. This results in one's toes sliding forward in the shoes and jamming in the front part of them (toe box). With a midfoot strike, any time the foot is on the ground, it should be pushing down and back to propel the body forward.

The Achilles tendon also plays a role in foot-contact time and therefore one's running efficiency. From a biomechanical perspective, the Achilles tendon is a spring that loads and unloads energy. The stiffer and longer the tendon, the greater the rebounding effect of the "spring" will be (496). While the length of the tendon is predetermined at birth, the stiffness of the tendon can be improved through strength training, running, and any other activity that stresses the Achilles.

Lastly, a study that observed the foot strike of 286 marathon runners found that when comparing the foot-strike type at 10K (6.2 miles) with 32K (20 miles), a large number of runners who ran with a midfoot strike at the 10K point converted to a heel strike at 32K (324). While not noted in the study, it could be hypothesized that fatigue and a slower pace were potential reasons for the change in foot-strike type.

FORM ANALYSIS

Recording a client on video while running on a treadmill for immediate feedback as well for reassessment purposes is of substantial value. Video of running on a treadmill is especially valuable for assessing foot strike, foot movement (pronation, supination, etc.), and overall body position.

While you can assess transverse hip rotation on the treadmill, it is often more accurately assessed outdoors, as the treadmill may facilitate hip rotation. The easiest way to assess runners outdoors is to either run alongside them or have them run on a track or back and forth in front of you. If running with your client, make sure that the road ahead of you is clear before looking at your client to the side. Video is a good medium for this as you can record the client running past you. You can also record your client running while you ride a bike beside the person. If filming while riding a bike, you must make sure that you are looking forward and in control of the bike at all times.

❖ STRIDE RATE AND LENGTH

Stride length and stride rate go hand in hand because of their effect on each other. The only way to increase running speed is to increase stride rate and/or stride length (568). Therefore a commonly asked question is, how long and how fast should one's stride be?

This is the million-dollar question with unfortunately no clear answer. Before we delve too deep into this question, the most important thing to appreciate is that like most aspects of exercise science, stride rate and length function on a bell curve. Too slow or too fast of a stride rate and too long or too short of a stride length will negatively impact a runner (573). Therefore a runner is most efficient somewhere in between the two extremes.

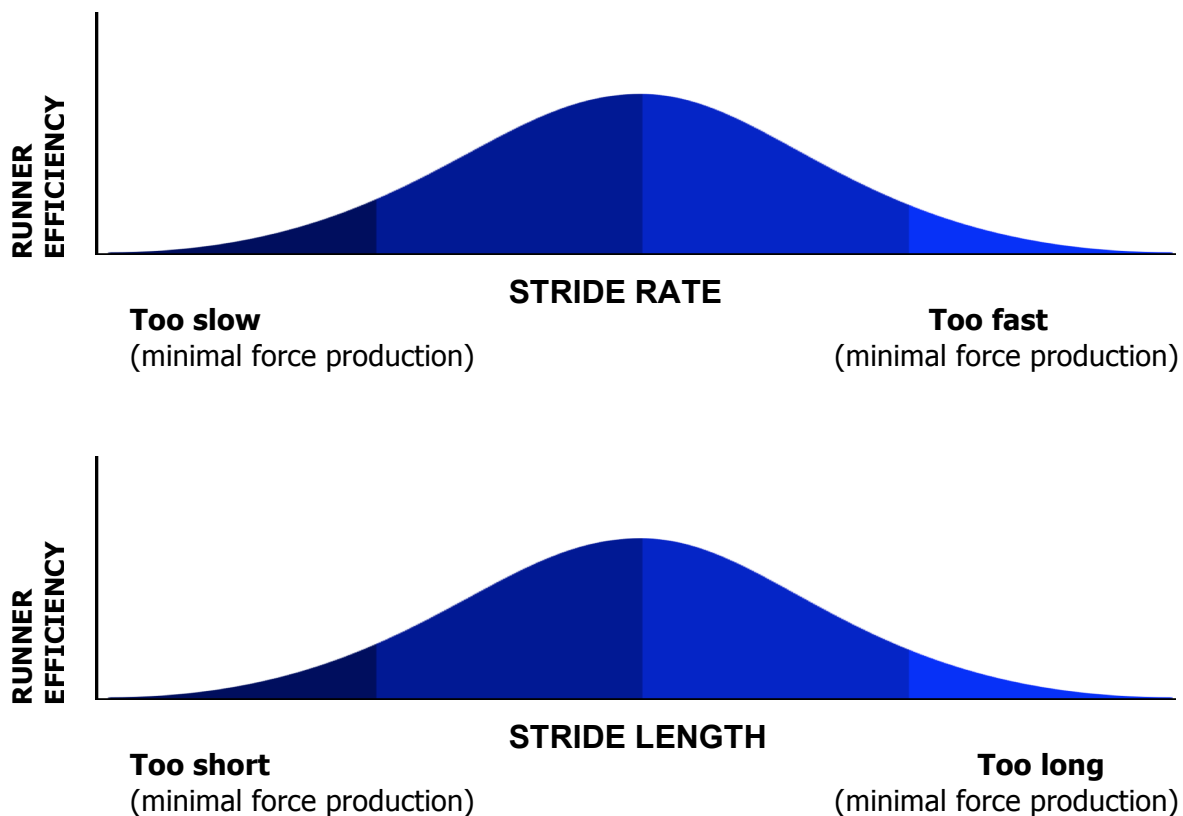


Figure 7.21 Stride length/rate curve

Argument against Changing Stride Length/Type

Two studies examined whether or not changing foot-strike type and stride length would increase performance or, more to the point, running economy (724, 725). Both studies found that changes in the stride did not improve running economy but rather had the opposite effect. So is changing one's stride a bad idea?

Not necessarily. The studies noted here occurred over a relatively short period. This is in contrast with the time likely required for neuromuscular adaptations to occur that would make a runner more efficient. If an individual has been running with a set stride for a long time, it almost guarantees that when initially switching strides, the person would have a decrease in running economy. Changes in biomechanical form require a long neuromuscular adaptation period.

If a runner is injury-free and appears to have good running economy, there is likely no reason to change form. However, if a client is injured, or has a form that is "less than ideal" and you believe the issue is biomechanical, working to change the stride might be the correct way to proceed.

CORRELATION BETWEEN STRIDE RATE AND STRIDE LENGTH

Initial increases in speed rely more on an increase in stride length versus stride rate. However, at higher running speeds, speed increases are more dependent on increasing stride rate and slightly decreasing stride length (569).

Interestingly, variables such as leg length, height, and leg mass (girth) are not primary factors in determining one's preferred stride rate and length (570).

Within an acceptable range, it is more energy efficient to lengthen the stride than to increase the stride rate (571). As heel strikers typically have a longer stride length than midfoot strikers, this may explain why toward the end of races, there is often a conversion from individuals striking with the midfoot to the heel (572).

STRIDE LENGTH

An individual's stride length is typically self-selected based on running speed. Some studies suggest that individuals self-select their running style (stride rate/length) based on what is most efficient and when forced to change their running gait, they become less efficient (574). This is likely due to an individual adapting to a particular running gait versus it being mechanically and physiologically most efficient. Typically when an individual changes established form, there will be an initial decrease in performance and efficiency until the body adapts to the new form.

To illustrate this point, watch a running race and you will see all sorts of stride lengths and rates – many of which are inefficient and not economical. If the theory that people self-select the ideal, and therefore most efficient, running gait was correct, all runners would be efficient in regard to their gait characteristics (foot strike, stride rate/length). This is not true.

The terms *step* and *stride* are often used interchangeably. However, they are actually two different things:

Step: Initial foot contact of one foot to the initial foot contact of the opposing foot

- Ex: Heel of left foot → heel of the right foot

Stride: Comprises two consecutive steps

- Ex: Heel of left foot → heel of right foot → heel of left foot

Therefore a stride is equal to two steps.

While running with a longer stride rate has been shown to be more economical than using a shorter stride rate from a cardiovascular standpoint, there is a limit to the stride length before it becomes inefficient. If this weren't the case, the gait of elite runners would be more akin to bounding than running.

Much of the efficiency of a runner has to do with the stretch-shortening cycle (SSC). It is theorized that too short of a stride minimizes the SSC. This equates to a small amount of potential energy that is stored (think: rubber band that is barely stretched) and used. Conversely, individuals who have a long stride (e.g., overstriders) store more potential energy than those with an overly short stride.

However, there is a reduction in the ability to use this potential stored energy (575).

The one caveat to running with an extremely short posterior stride occurs when running up a steep incline on a rough or unpaved surface. In these cases, a short posterior stride is needed to reduce the chance for the drive leg slipping on loose surfaces because of too much leg/hip extension.

In summation, the “perfect” stride length should be long enough to store an adequate amount of elastic energy and short enough to be able to utilize it. This typically relates to a short anterior stride length (foot hitting ground directly in front of body or directly under body) and a relatively long posterior stride to take advantage of proper hip extension and transverse rotation.



Video 7.7 Stride Length

Vertical Oscillation

An aspect of stride length often not discussed is that of vertical oscillation. This pertains to the vertical movement of the body while running. From an efficiency standpoint, it takes much more energy to move the body vertically than it does horizontally. Therefore it is suggested to run with as little vertical oscillation as possible. Too much vertical oscillation results in a runner “bounding” or “bouncing.” Conversely, a runner with minimal vertical oscillation typically shuffles. Therefore the ideal amount of vertical oscillation exists on a bell curve. Additionally, greater vertical oscillation equates to higher ground forces (i.e., impact on body).

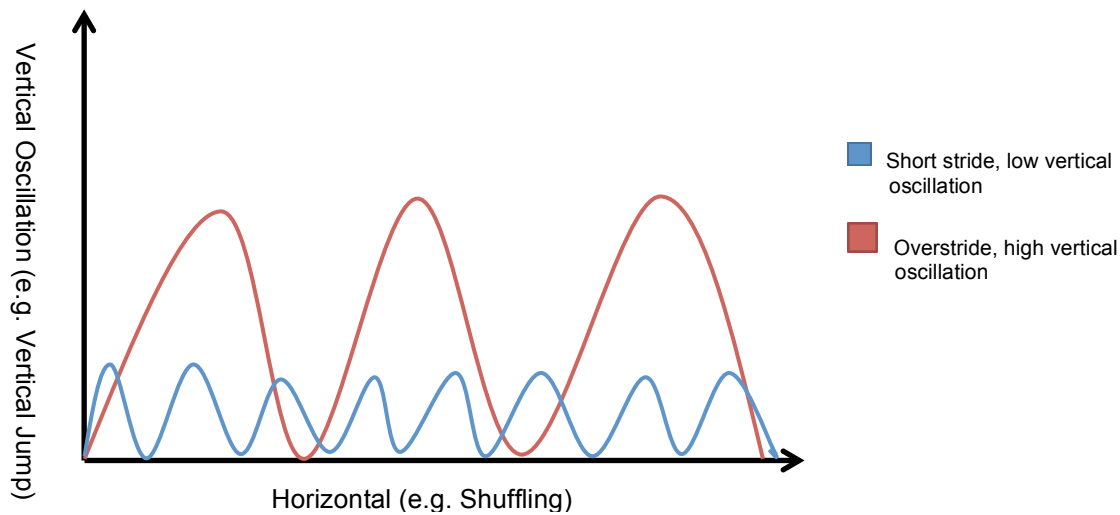


Figure 7.22 Vertical Oscillation

In regard to stride length, the longer the stride, typically the greater the vertical oscillation. An analogy to this is throwing a ball. In order to throw a ball far (e.g., long stride), the ball must have a substantial arc on it (e.g., a lot of vertical oscillation).

In addition to a shorter stride length, transverse rotation of the pelvis (hip rotation) assists in reducing vertical oscillation due to the energy being used for forward versus vertical propulsion.

The primary determinant regarding the amount of vertical oscillation pertains to how the drive leg pushes against the ground. The correct angle of force is downward and rearward (diagonally rearward). When a runner pushes directly down into the ground, the body moves primarily upward and not forward.



Video 7.8 Vertical Oscillation

BACKPACK ASSESSMENT

Running with a lightly loaded backpack is a great way to assess vertical oscillation, as well as torso rotation. While it's not needed (or enjoyable) to run with a loosely worn backpack for long distances, doing so for a short distance (200–400 meters) is a good way to assess one's form. Load a backpack with a

few T-shirts, and any extra support straps on the pack across the chest or waist should not be fastened.

When running with the proper amount of vertical oscillation, the pack should slightly move up and down. If the pack moves up and down more than a few inches while running, it's likely that the runner has too much vertical oscillation. This provides instantaneous feedback to clients and thus allows them to adjust their form on the fly to reduce vertical oscillation.

As an ancillary benefit, this assessment can also be used to determine if there is a lack of hip/torso rotation. If the pack does not sway from left to right but only moves vertically, it is likely that the client does not have sufficient hip/torso rotation.

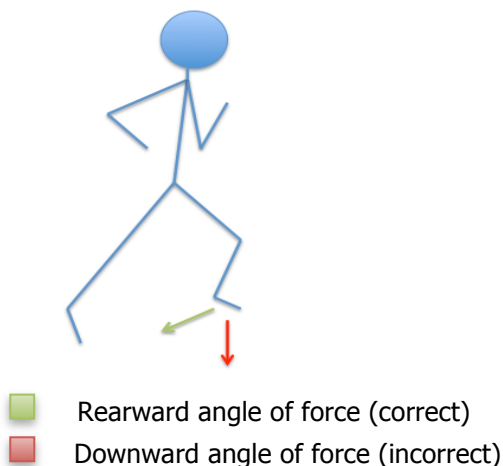


Figure 7.23 Force direction

In the preceding image, the green arrow represents the correct angle of force whereas the red arrow represents the incorrect angle.

STRIDE RATE

Stride rate (SR) directly affects stride length (SL) and vice versa. The topic of stride rate is often discussed from the perspective that a fast stride rate is better than a slow stride rate. Two things to consider:

1. Running with a fast stride rate does not necessarily equate to having proper form or being efficient.
2. What constitutes a “fast” SR is subjective and largely based on the speed of the runner.

Running with a high SR is typically indicative of running at a relatively high rate of speed. Therefore increasing a runner’s SR without increasing speed will likely reduce efficiency. Increasing one’s SL is largely a function of increased force production whereas increasing SR is likely due to increased neuromuscular and cardiovascular capacity (578). Therefore it is probable that a runner who is currently running with a low SR cannot increase SR dramatically without a decrease in form because of inadequate neural connections and cardiovascular fitness.

At the 1984 Olympics, renowned running coach Jack Daniels observed the stride rates of runners and found that all but one runner had a stride rate of 180 or more per minute (576). This data point has found its way into running folklore as being representative of the low end of what constitutes a “fast” or “high” stride rate. Therefore an SR of 180 has become a benchmark for many runners who are looking to increase their SR and overall running performance.

As SR is largely influenced by a runner’s speed, it is not surprising that Olympic-caliber runners have high SRs, even for long-distance events (576). However, as most runners do not have the leg turnover of Usain Bolt, it is expected that the SR of a good majority of runners will be below 180. Additionally, a runner who attains an SR of 180 on a fast training workout may likely be below 180 on a long run.

As noted by running coach Steve Magness, if an individual runs at an 11-minute/mile pace with an SR of 180, this equates to a stride length of only 32 inches – hardly an efficient SL (577)!

CONCLUSIONS

Decreasing SL and increasing SR are common guidelines to increase running performance. However, any changes in SL and/or SR must be done with a focus on proper form. While these guidelines are not incorrect per se, they must be appreciated in the context of a runner as a whole. As noted previously, any

change in form typically equates to a drop in performance initially because of adaptation process.

It is natural for SR and SL to change during a race or workout (577, 578). This is especially the case when fatigue sets in (577).

An interesting point of view regarding SL and SR comes from Steve Magness, who views these two data points as feedback. In other words, focus on the other elements of form (foot strike, hip rotation, upper-body position, etc.) and the SL and SR will change accordingly (577).

Lastly, as a coach, do not become fixated on a set number regarding the optimal SR and SL. These are influenced on an individual level, and therefore there is no right or wrong SR and SL numbers.

❖ BIOMECHANICAL RUNNING ASSESSMENTS

These assessments should be evaluated while standing (static) and running (dynamic).

KNEES

Issue: Valgus knees (knock-knees)

Potential Muscular/Structural Causes:

- Underactive glute maximus/medius
- Overactive hip adductors

Potential Structural Issues:

- Foot arch collapses, resulting in excessive pronation
- Excessive Q angle

Potential Equipment Issues: shoes with lack of proper arch/heel support

Action: STRETCH: hip adductors

STRENGTHEN: glute max/medius, intrinsic foot muscles, calf muscles

Issue: Varus knees (bowlegged)

Potential Structural Issues:

- Slight appearance of a varus knee position is often the result of foot supination.
- Excessive varus knees can also be caused by a structural deformity of the femur and tibia/fibula. This is called *genu varum* and is typically rectified only via surgery or bracing.

BACK

Issue: Excessive lumbar/thoracic extension (excessive anterior pelvic tilt)

Potential Muscular/Structural Causes:

- Overactive erector spinae, hip flexors
- Underactive abdominals

Action: STRETCH: erector spinae, mid/low trapezius, hip flexors

STRENGTHEN: rectus abdominis, hamstrings, glutes

Issue: Excessive lumbar/thoracic flexion (excessive posterior pelvic tilt)

Potential Muscular/Structural Causes:

- Overactive abdominals
- Underactive erector spinae

Action: STRETCH: rectus abdominis, hamstrings

STRENGTHEN: erector spinae

SHOULDERS

Issue: Shoulders elevate and internally rotate

Potential Muscular/Structural Causes:

- Overactive upper trapezius
- Underactive lower trapezius and rhomboids

Action: STRETCH: upper trapezius, pectoralis major, subscapularis

STRENGTHEN: rear deltoid, rhomboids, mid/low trapezius, infraspinatus, teres minor

HIPS

Issue: Pelvic tilt in frontal plane*

Potential Muscular/Structural Causes:

- Weak hip abductors (glute medius)
 - Termed *Trendelenburg gait*, this biomechanical issue causes the contralateral side of the pelvis to tilt downward because

of a lack of hip abductor strength on the ipsilateral side of the pelvis.

- Spinal scoliosis
- Leg-length discrepancy
- Uneven foot support (arch)

Issue: Rotated (twisted) pelvis*

Potential Muscular/Structural Causes:

- Uneven calf tightness
 - Leg-length discrepancy
 - Uneven foot support (arch)
 - Sacroiliac joint injury
 - Muscle imbalance of erector spinae and quadratus lumborum
-
- If your client presents with a tilted (frontal plane) and/or rotated pelvis (transverse), it is advised to refer the individual to a clinical specialist because of the complexity of the issue.

❖ SUMMARY

- The three phases of running gait are:
 - Drive
 - Support
 - Recovery
- Mild pronation is normal. Supination and overpronation are considered abnormal foot/gait patterns.
- Ground reaction force relates to the force acted upon the body in relation to the force of the foot strike on the ground.
- Pronation is not predictive of pain or injury.
- Q angle relates to the angle of the femur.
- Utilization of free energy reduces the active muscle requirement while still producing the required amount of force.
- The foot should strike the ground under the runner with the tibia at or close to 90 degrees in relation to the ground during the foot strike.
- The majority of energy when running goes to supporting the body.
- The hips should rotate slightly in the transverse plane when running. This is a source of energy termed *the spinal engine*.
- The PSSS and ASSS systems utilize the stretch-shortening cycle to store potential energy and use kinetic energy.
- Excessive vertical body movement equates to wasted energy.
- Generally, the faster one runs, the foot strike occurs farther forward on the foot.
- A heel strike places more stress on the skeletal system whereas a midfoot strike places more stress on the musculature system.
- Running with a short stride and fast turnover is often preferable to running with a long stride and a slow turnover.